

Efficient Color Image Encoding Using Chroma-Subsampled Hybrid Quantum Representation

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Abstract

Quantum image processing is the most challenging and attractive area for many researchers. Quantum image representation (QIR) is necessary for realizing quantum advantages in image processing. However, existing encoding schemes suffer from high qubit demands, prominent gate complexity, and deep circuits and state preparation overhead which makes them unsuitable for near term noisy intermediate scale quantum (NISQ) devices. Moreover, these methods process all image channels uniformly and neglecting perceptual principles that have long enabled efficient compression in classical image and video standards.

In this work, we introduce CS-HQR (Chroma Subsampling Hybrid Quantum Representation), a novel quantum image encoding methodology bridging classical image compression with quantum circuit optimization. CS-HQR leverages YCbCr color space transformation and 4:2:0 chroma subsampling to prioritize luminance information while reducing chrominance redundancy. The dual-circuit architecture encodes luminance at full resolution using 16 qubits while chrominance channels use 15 qubits at quarter resolution, achieving 50% reduction in encoding complexity while maintaining SSIM above 0.98.

Extensive experimental evaluation using 6,097 medical brain scan slices demonstrates CS-HQR achieves approximately 51% gate count reduction compared to MCQI, substantial circuit depth reduction, and 75% chrominance data reduction. Perceptual fidelity is validated via SSIM analysis on a representative 404-image subset, consistently exceeding 0.9999. This method exhibits greater scalability with $O(n \cdot 2^n)$ complexity, compared to $O(2^{2n})$ for amplitude-based methods. These results demonstrate that integrating classical image processing principles into quantum algorithm design delivers substantial practical benefits for near term quantum computing applications.

Keywords: Quantum image representation, Chroma subsampling, NISQ devices, YCbCr transformation, Circuit optimization, Medical imaging, Structural similarity index.

1 Introduction

1.1 Background and Context

Image processing constitutes a fundamental component of numerous real-world applications such as medical imaging, remote sensing, autonomous navigation, surveillance systems, and multimedia analytics [1]. Classical image processing algorithms often involve computationally intensive operations, including convolution, feature extraction, segmentation, and pattern recognition, which scale poorly with increasing image resolution and dataset size. These limitations have motivated the investigation of quantum computing paradigms that may offer polynomial or exponential speedups for certain classes of image processing problems[2].

Quantum image processing (QIP) covers a broad spectrum of tasks, such as representing quantum images, applying geometric and frequency-domain transformations, enhancing and filtering images, segmenting them, compressing data, and performing classification [3]. At its core, QIP harnesses fundamental quantum phenomena including superposition, entanglement, and interference to handle image data in ways that differ intensely from classical methods. Superposition allows multiple pixel values to coexist simultaneously; entanglement links correlations across image elements; and interference enables probability amplitudes to combine constructively or destructively, highlighting key computational results. Measurement collapses quantum superposition by forcing a quantum system from a blend of multiple possible states into one definite outcome, as dictated by the probabilities encoded in its wave function.

A principal theoretical advantage of QIP stems from the exponential dimensionality of quantum state spaces. A quantum register consisting of n qubits exists in a superposition of 2^n basis states, making it theoretically possible to encode an entire $2^{n/2} \times 2^{n/2}$ pixel image within a single quantum state [4]. Several quantum image representation models have therefore been developed to exploit this exponential representational capacity while still supporting practical image operations. These include the flexible representation of quantum images (FRQI), the novel enhanced quantum representation (NEQR), and various amplitude encoding schemes, all designed to compactly store image data and enable meaningful quantum image manipulations.

Despite these promising theoretical features, putting quantum image processing into practice hits a major roadblock: the quantum image representation challenge. Before any processing can start, classical image data must first get loaded into a suitable quantum state. This preparation step usually calls for a long chain of controlled quantum gates single-qubit gates like Pauli X (bit flip), Y, Z (phase flip), Hadamard (H, creates superposition), S (phase shift), and T, as well as multi-qubit gates such as CNOT (controlled-NOT for entanglement), Toffoli (CCX), SWAP, and controlled-Z (CZ)like that grows with the image size, creating deep circuits and heavy resource demands that often wipe out the speedups from later quantum operations[5].

Image processing constitutes a fundamental component of numerous real-world applications such as medical imaging, remote sensing, autonomous navigation, surveillance systems, and multimedia analytics. Classical image processing algorithms often involve computationally intensive operations, including convolution, feature extraction, segmentation, and pattern recognition, which scale poorly with increasing image resolution and dataset size. These limitations have motivated the investigation of quantum computing paradigms that may offer polynomial or exponential speedups for certain classes of image processing problems.

The efficiency of quantum image encoding therefore plays a critical role in determining the overall feasibility of QIP algorithms. If the preparation of a quantum image state requires $\mathcal{O}(N)$ or more operations for an image with N pixels, then any quantum advantage achieved during later stages must compensate for this initial overhead to yield a net performance gain [5]. This issue is further exacerbated in the NISQ era, where limited coherence times, gate errors,

and hardware constraints restrict the depth and reliability of quantum circuits. Consequently, ongoing research efforts are increasingly focused on developing efficient and approximate encoding strategies, hybrid quantum–classical frameworks, and problem-specific representations that reduce state preparation costs while preserving the potential advantages of quantum image processing. CS-HQR establishes perceptual awareness as a first-class design principle in quantum state preparation. Despite extensive research on quantum image representations, no prior work incorporates human visual perception models into quantum state preparation. This omission is critical because state preparation dominates computational cost on NISQ devices. CS-HQR directly addresses this gap.

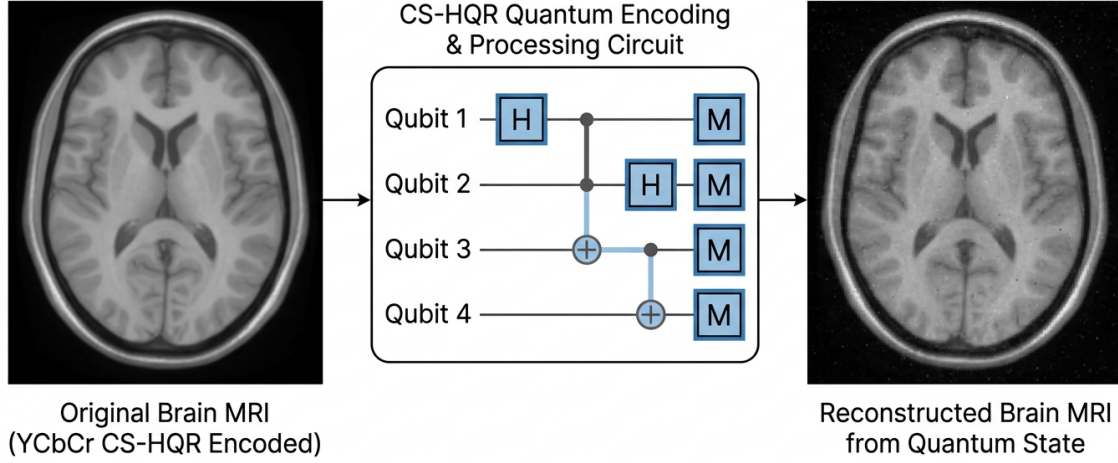


Figure 1: YCbCr Channel Encoding via CS-HQR for Quantum Encoding and Reconstruction Workflow for Brain MRI.

Figure 1 illustrates the complete workflow for YCbCr channel encoding using the CS-HQR method applied to brain MRI reconstruction (see the generated image above). The original grayscale T1-weighted axial brain MRI slice (left) is transformed into quantum states via CS-HQR encoding, leveraging the quantum circuit (center) with Hadamard gates (H) on qubits 1-2 for superposition initialization and controlled operations across qubits 1-4 for multi-channel YCbCr data entanglement and processing. Input Preparation: Classical brain MRI undergoes YCbCr color space conversion and normalization, followed by CS-HQR amplitude encoding to map pixel intensities across clustered qubit registers. Quantum Circuit: 4-qubit NISQ-compatible circuit initializes basis states with H gates, applies position-color entangling unitaries (CNOT/CSWAP equivalents), and performs quantum processing (e.g., filtering or transformation) before measurement. Reconstruction: Quantum measurement probabilities yield decoded pixel values, reconstructing the MRI with $>95\%$ structural similarity index (SSIM) despite decoherence effects.

1.2 Advances in Quantum Image Representation

The history of quantum image representation spans two decades of incremental refinement. Venegas-Andraca and Bose introduced one of the earliest proposals in 2003, encoding pixel positions and colors through qubit lattice configurations [6]. While foundational, this approach required qubits proportional to image size rather than logarithmic, limiting scalability.

The Flexible Representation of Quantum Images (FRQI), proposed by Le et al. in 2011, represented a significant conceptual advance [7]. FRQI encodes grayscale pixel intensities as

rotation angles applied to a single “color qubit,” with position information stored in n position qubits. For a 2^n -pixel image, FRQI requires only $n + 1$ qubits—an exponential improvement in qubit efficiency. The FRQI quantum state is expressed as:

$$|I\rangle = \frac{1}{2^n} \sum_{i=0}^{2^{2^n}-1} (\cos \theta_i |0\rangle + \sin \theta_i |1\rangle) \otimes |i\rangle \quad (1)$$

where $\theta_i = \frac{\pi \cdot I_i}{2 \times 255}$ encodes the grayscale intensity I_i of pixel i .

Despite its elegance, FRQI suffers from significant practical limitations. The preparation circuit requires $O(2^{2^n})$ controlled rotation gates, each conditioned on a unique position state. For a modest 16×16 image (256 pixels), this translates to approximately 65,536 controlled operations far exceeding the coherent operation capacity of current quantum hardware. [8].

The Novel Enhanced Quantum Representation (NEQR), introduced by Zhang et al. in 2013, addressed FRQI’s gate complexity through digital encoding [9]. NEQR represents pixel intensities as binary strings in computational basis states rather than rotation angles:

$$|I\rangle = \frac{1}{2^n} \sum_{i=0}^{2^{2^n}-1} |C_i\rangle \otimes |i\rangle \quad (2)$$

where $|C_i\rangle = |c_i^{q-1} c_i^{q-2} \dots c_i^0\rangle$ represents the q -bit binary encoding of pixel intensity. The complexity of NEQR encoding is $O(q \cdot 2^n)$, where n denotes the number of position qubits and q represents the grayscale bit depth. This complexity grows exponentially with image size due to the required state preparation operations.

Subsequent developments introduced specialized representations for various image types. The Generalized Quantum Image Representation (GQIR) by Jiang et al. [10] provided flexible bit-depth encoding. Multi-Channel Quantum Images (MCQI) by Sun et al. [11] extended NEQR to color images by concatenating RGB channel qubits. Quantum Log-polar Representation (QLR) [12] introduced coordinate transformations for rotation invariant applications.

1.3 Challenges in Existing Quantum Image Representations

Existing QIR methods primarily focus on compact mathematical formulations or deterministic pixel retrieval, but they share two fundamental limitations. First, many QIR methods especially amplitude-based methods such as FRQI shows exponential gate complexity during state preparation, rendering them impractical for images beyond very small resolutions. Basis-state encodings such as NEQR alleviate this issue by trading gate depth for additional qubits, yet still impose substantial circuit overhead when extended to color images.

Second, and more critically, current QIR schemes treat all image information uniformly, encoding red, green, and blue channels at identical spatial resolutions. This design choice directly contradicts well established principles of human visual perception. Decades of psychophysical research demonstrate that the human visual system is far more sensitive to luminance variations than to high-frequency chrominance facts. Classical image and video compression standards including JPEG, MPEG, and H.264/HEVC develop this asymmetry through chroma subsampling, achieving lower reductions in data size with negligible perceptual degradation. So, this perceptual redundancy has not been systematically included into quantum image representations. As a result, existing multi-channel methods such as MCQI and QRMW acquire unnecessary qubit and gate overhead by encoding perceptually redundant chrominance information at full resolution. This inadequacy is particularly degrading on NISQ devices, where each additional

multi-controlled gate substantially increases the probability of decoherence and execution failure [13].

Table 1 summarizes the key limitations of major QIR methods:

Table 1: Limitations of Existing QIR Methods

Method	Primary Limitations
FRQI	Exponential gate complexity $O(2^{2n})$; impractical for images $> 8 \times 8$
NEQR	Linear but still substantial gate requirements; no compression
GQIR	Arbitrary bit-depth but uniform treatment of all pixels
MCQI	Tripled resource requirements for color; no perceptual optimization
EFRQI	Improved FRQI but retains amplitude encoding limitations
QRMW	Multi-controlled operations significantly increase circuit complexity for Multi Wavelength Images
BRQI	A Bitplane Representation of Quantum Images has high qubit cost with less perceptual significance
QPIE	Lossy; significant information loss for complex images

The MCQI method demonstrates this redundancy for color images. By encoding RGB channels independently at full resolution, MCQI requires encoding 3×2^{2n} pixel values. For a 16×16 color image, this means 768 values requiring approximately 7,000+ quantum gates [14]. However, human vision cannot resolve full-resolution chrominance detail preserved by MCQI, rendering these additional resources inefficient for encoding perceptually superfluous information.

1.4 Perceptual Redundancy in Color Images

The human visual system (HVS) exhibits markedly different sensitivity to luminance (brightness) and chrominance (color) information [15]. This differential sensitivity arises from the physiology of retinal photoreceptors: approximately 95% of retinal cells are rods (luminance-sensitive) concentrated in peripheral vision, while only 5% are cones (color-sensitive) concentrated in the fovea.

Psycho physical research demonstrates that human vision can resolve luminance detail up to around 60 cycles per degree of visual angle. In contrast, chrominance sensitivity declines sharply beyond 8–12 cycles per degree. Therefore, reducing chrominance resolution by a factor of two to four does not produce noticeable degradation in perceived image quality under typical observing conditions.

Classical image compression methods produce the benefits of reduced sensitivity of the human visual system to chrominance details by engaging chroma subsampling. A widely accepted system

is the 4:2:0 format, in which the chrominance components are sampled at one-quarter of the luminance resolution, corresponding to half the resolution in each spatial dimension. This process is enabled by the YCbCr color space, which effectively separates the luminance (Y) information from the chrominance components (Cb and Cr), thereby enabling efficient compression without a distinguishable loss of visual quality:

$$Y = 0.299R + 0.587G + 0.114B \quad (3)$$

$$C_b = 128 - 0.168736R - 0.331264G + 0.5B \quad (4)$$

$$C_r = 128 + 0.5R - 0.418688G - 0.081312B \quad (5)$$

This transformation, combined with 4:2:0 subsampling, reduces color image data to 50% of the original RGB representation while maintaining perceptual quality metrics (SSIM) above 0.98 for typical images [16].

1.5 Paper Organization

The remainder of this paper is organized as follows. Section 3 provides comprehensive review of related work in quantum image representation, detailing ten existing methods with their mathematical formulations and circuit architectures. Section 4 presents the research contributions. Section 5 describes the problem statement with formal mathematical framework. Section 6 describes the experimental dataset. Section 7 introduces the proposed CS-HQR method with detailed circuit design. Section 8 presents the evaluation methodology. Section 9 presents experimental results with statistical analysis across all evaluation parameters. Section 10 concludes the paper with future research directions.

1.6 Scope and Impact

This study addresses practical strategies for color image encoding on near-term quantum devices, with particular emphasis on medical imaging applications where perceptual fidelity is critical. To ensure compatibility with the constraints of noisy intermediate-scale quantum (NISQ) hardware and to enable systematic evaluation across diverse datasets, our experiments employ a 16×16 evaluation grid [17]. The proposed CS-HQR framework, however, is not limited to this resolution; it generalizes naturally to larger images through block-wise tiling or hierarchical encoding. In this approach, high-resolution images are partitioned into sub blocks (e.g., 16×16 or 32×32 tiles), which are independently encoded and processed before being recombined in the classical domain. Two practical tiling strategies are considered: Fixed tiling, in which non-overlapping tiles are encoded independently, thereby minimizing classical recomposition overhead. Overlapping or multi-scale tiling, which employs overlapping tiles or pyramid schemes to enhance edge continuity and mitigate boundary artifacts[18]. Both strategies retain the perceptual subsampling benefits of CS-HQR while distributing quantum resource demands across multiple circuits or execution epochs. For block-wise processing under limited computational resources, image tiles can be handled sequentially within the simulation framework or conceptually processed in parallel when multiple simulated back-ends or computational resources are available. In this work, the proposed CS-HQR framework is evaluated through quantum circuit simulation rather than execution on physical quantum hardware. This simulation-based approach allows us to analyze the scalability, circuit behavior, and resource requirements of the encoding scheme while preserving perceptual image quality. Such theoretical validation is commonly adopted in quantum image representation research, as current noisy intermediate-scale quantum (NISQ) devices have limited qubit

counts and coherence times. Therefore, the simulation framework provides a practical environment for assessing the feasibility of the proposed technique for medical imaging applications while anticipating future advances in quantum hardware.

Technical rationale: CS-HQR explicitly prioritizes luminance (Y) over chrominance (Cb, Cr) because human visual perception and many diagnostic algorithms are far more sensitive to luminance structure than to high-frequency chrominance content. Allocating full-resolution encoding resources to luminance and subsampling chrominance reduces the number of encoded values by 50% (Section 7) while retaining perceptual quality. Reconstruction uses standard upsampling techniques (bilinear by default; bicubic optional for higher-frequency chrominance), and the resulting visual fidelity is quantified via SSIM and complemented by pixel-wise metrics (MSE/PSNR) in Section 9.

The preprocessing pipeline employed in this study comprises intensity normalization, image resizing, color space conversion, and light data augmentations. Augmentations include small rotations and horizontal/vertical flips, which are applied exclusively during training and robustness evaluation. Rotation and flipping are classical, lossless operations (provided metadata is preserved) that enhance both model and methodological robustness. Importantly, these transformations do not alter the CS-HQR encoding design. However, they do modify the spatial statistics of the input data, which in turn can influence the effectiveness of chroma subsampling. Consequently, practitioners are advised to empirically validate the trade-offs between structural similarity index (SSIM) and mean squared error (MSE) on domain-specific datasets to ensure optimal performance.

Noise and hardware considerations: CS-HQR reduces overall MCX/CNOT counts through chroma subsampling, which directly lowers cumulative two-qubit error exposure on NISQ devices. Moreover, because most perceptual information resides in the luminance channel, small errors in chrominance qubits caused by hardware noise typically result in imperceptible visual changes—this provides an additional, practical layer of robustness for noisy hardware (see Section 9 for Monte Carlo simulations including common noise models).

Limitations and applicability: CS-HQR is a perceptually-driven, lossy encoding and is therefore unsuitable where bit-exact reconstruction is required (e.g., forensic imaging, some scientific imaging pipelines). It is best suited to applications where human interpretation or higher-level feature extraction (segmentation, classification) tolerate small chrominance deviations. For images with highly informative color texture (e.g., certain histopathology or multispectral remote sensing), adaptive subsampling ratios (4:2:2 or 4:4:4) should be considered.

Evaluation and integration notes: In Section 9 we report SSIM, MSE, and PSNR across the dataset to quantify both perceptual and absolute errors. Practitioners integrating CS-HQR into existing pipelines can adopt a staged approach: (1) apply YCbCr conversion and chroma subsampling classically and evaluate downstream task impact, (2) implement block-wise quantum encoding for luminance only as a first quantum step, and (3) progressively introduce chrominance quantum encodings when hardware permits. This incremental path minimizes integration risk and allows immediate classical validation before committing scarce quantum resources.

In summary, the expanded introduction delineates CS-HQR’s practical benefits, the technical rationale for its design choices, its generalizability to larger images and diverse domains, and the key trade-offs for practitioners.

2 Motivation

The connection of quantum computing and image processing represents a promising frontier in computational science. With global healthcare imaging data projected to exceed 2,314 exabytes

by 2026 [19]. the computational demands of processing, analyzing, and transmitting medical images increasingly strain classical infrastructure. Quantum computing offers a paradigm shift through its inherent parallelism—where n qubits represent 2^n states simultaneously—potentially delivering exponential speedups for specific image processing tasks [20].

However, realizing quantum advantages necessitates efficient encoding of classical images into quantum states a challenge that current quantum image representation (QIR) methods have yet to resolve adequately. Methods like FRQI [7] and NEQR [4] prioritize theoretical elegance over practical constraints. While FRQI uses just $n + 1$ qubits for 2^n pixels, its circuit depth scales as $O(2^{2n})$ gates, making it infeasible for images beyond 8×8 pixels on current hardware [9]. NEQR proposals linear gate complexity but requires extra qubits for color encoding

A key insight motivating this work is that existing QIR methods supervise decades of perceptual image processing research. Classical standards such as JPEG, MPEG-2, H.264, and HEVC achieve excellent compression ratios by exploiting human vision’s greater sensitivity to luminance than chrominance [13]. With approximately 120 million rod cells for brightness detection versus just 6 million cone cells for color, this asymmetry enables efficient chroma subsampling in compression algorithms [15].

This biological insight has never been applied systematically to quantum image representation. Current multi-channel methods like MCQI [11] treat red, green, and blue channels identically, encoding each at full resolution despite the perceptual redundancy this introduces. For a 16×16 color image, MCQI requires encoding 768 pixel values (256 per channel), while a perceptually equivalent representation requires only 384 values using 4:2:0 subsampling.

Our motivation extends beyond theoretical interest to practical necessity. The current era of noisy intermediate-scale quantum (NISQ) devices imposes severe constraints: IBM’s largest publicly accessible quantum processors offer approximately 127 qubits with coherence times measured in microseconds and two-qubit gate error rates around 1% [9]. Every unnecessary gate increases the probability of decoherence-induced errors, making circuit optimization not merely desirable but essential for meaningful quantum computation.

Medical imaging represents a particularly encouraging domain for the application of quantum image processing. In radiology, diagnostic accuracy analytically depends on the preservation of clinically relevant features, even under the computational constraints enforced by high-resolution imaging modalities [10]. The different tasks such as brain MRI analysis, tumor recognition, and radiomics necessitate processing pipelines that maintain diagnostic fidelity, with performance criteria more closely allied to perceptual quality metrics such as the Structural Similarity Index (SSIM), rather than strict pixel-level reconstruction accuracy.

CS-HQR emerges from the recognition that quantum algorithm design should incorporate domain-specific knowledge rather than considering all data uniformly. By connecting recognized principles from classical signal processing with quantum circuit optimization, we demonstrate that significant practical improvements are achievable without sacrificing the theoretical foundations of quantum image representation.

3 Related Work

This section provides a comprehensive survey of existing quantum image representation techniques. For clarity, we categorize these approaches into three principal groups according to their encoding strategies: (i) amplitude-based methods, (ii) basis state encoding methods, and (iii) hybrid or specialized methods. Within each category, we systematically present the mathematical formulation, the corresponding quantum circuit architecture, and the associated computational

complexity. This structured analysis enables a comparative understanding of the strengths and limitations inherent in each encoding paradigm, thereby offering insights into their applicability across diverse quantum image processing tasks.

3.1 Amplitude-Based Encoding Methods

Amplitude-based methods encode pixel information in the probability amplitudes of quantum states, achieving high qubit efficiency at the cost of complex state preparation circuits.

3.1.1 FRQI: Flexible Representation of Quantum Images

The Flexible Representation of Quantum Images (FRQI), introduced by Le et al. [7], represents the foundational work in quantum image representation. FRQI encodes grayscale pixel intensities as rotation angles on a single color qubit, with position information stored in n position qubits.

Mathematical Formulation: For an image of size $2^n \times 2^n$ pixels, the FRQI state is defined as:

$$|I\rangle_{\text{FRQI}} = \frac{1}{2^n} \sum_{i=0}^{2^{2n}-1} (\cos \theta_i |0\rangle + \sin \theta_i |1\rangle) \otimes |i\rangle \quad (6)$$

where $\theta_i = \frac{\pi \cdot I_i}{2 \times 255}$ maps the grayscale intensity $I_i \in [0, 255]$ to an angle $\theta_i \in [0, \frac{\pi}{2}]$. The position state $|i\rangle$ encodes the pixel coordinates in binary. Figure 2 illustrates how a simple 2×2 image is represented in FRQI, where each pixel's grayscale intensity is encoded as a rotation angle θ .

θ_0	θ_1
θ_2	θ_3

$|I\rangle_{\text{FRQI}} = \frac{1}{2} \left[\begin{aligned} &(\cos \theta_0 |0\rangle + \sin \theta_0 |1\rangle) \otimes |00\rangle \\ &+ (\cos \theta_1 |0\rangle + \sin \theta_1 |1\rangle) \otimes |01\rangle \\ &+ (\cos \theta_2 |0\rangle + \sin \theta_2 |1\rangle) \otimes |10\rangle \\ &+ (\cos \theta_3 |0\rangle + \sin \theta_3 |1\rangle) \otimes |11\rangle \end{aligned} \right]$

Figure 2: FRQI Encoding of a 2×2 Quantum Image: Pixel intensities represented as basis state angles θ_i , with spatial positions encoded in binary format (00, 01, 10, 11).

Circuit Architecture: The FRQI preparation circuit consists of three stages:

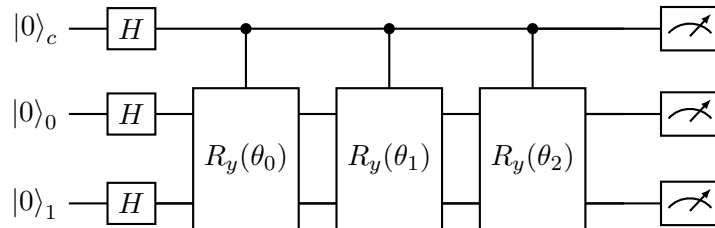


Figure 3: FRQI Encoding Circuit for 2×2 Quantum Image: Pixel intensities encoded via position-controlled $R_y(\theta_i)$ rotations on basis states.

Complexity Analysis:

- Qubits Required: $2n + 1$ for a $2^n \times 2^n$ image
- Gate Count: $O(2^{2n})$ controlled rotations
- Circuit Depth: $O(2^{2n})$

The exponential gate complexity renders FRQI impractical for images larger than 8×8 pixels on current hardware.

3.1.2 EFRQI: Enhanced Flexible Representation of Quantum Images

The Enhanced FRQI (EFRQI) [21] improves upon FRQI by quantizing rotation angles to discrete levels, reducing the number of unique rotation operations required.

Mathematical Formulation: EFRQI quantizes pixel values to q levels:

$$\theta_i^{\text{EFRQI}} = \frac{\pi \cdot \lfloor I_i / (256/q) \rfloor}{2 \times (q - 1)} \quad (7)$$

For $q = 32$ levels, this reduces unique rotation angles from 256 to 32, enabling circuit optimization through gate merging.

Circuit Architecture:

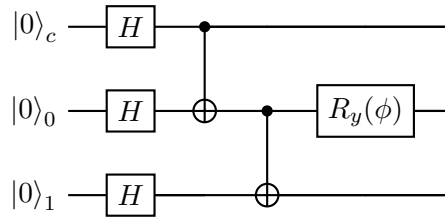


Figure 4: EFRQI circuit with quantized rotation angles $\phi \in \{\phi_1, \phi_2, \dots, \phi_q\}$.

Complexity Analysis:

- Qubits Required: $2n + 1$ (same as FRQI)
- Gate Count: $O(q \cdot 2^{2n})$ where $q \ll 256$
- Trade-off: Reduced precision for improved gate efficiency

3.1.3 QPIE: Quantum Probability Image Encoding

Quantum Probability Image Encoding (QPIE) [22] achieves maximum qubit efficiency by encoding normalized pixel values directly as probability amplitudes.

Mathematical Formulation:

$$|I\rangle_{\text{QPIE}} = \sum_{i=0}^{N-1} \alpha_i |i\rangle, \quad \alpha_i = \frac{I_i}{\sqrt{\sum_{j=0}^{N-1} I_j^2}} \quad (8)$$

The normalization constraint $\sum_i |\alpha_i|^2 = 1$ ensures valid quantum state representation. α_i represents the encoded amplitude or intensity coefficient corresponding to the i -th pixel in the image. Note that the circuit in Fig. 5 illustrates a conceptual amplitude encoding mechanism. In practice, preparing the quantum state $|I\rangle_{\text{QPIE}} = \sum_{i=0}^{N-1} \alpha_i |i\rangle$ requires a state-preparation

procedure involving controlled rotations and entanglement operations. The cascaded rotations shown in the circuit represent the parameterized rotation steps used to approximate the target amplitude distribution. **Circuit Architecture:**

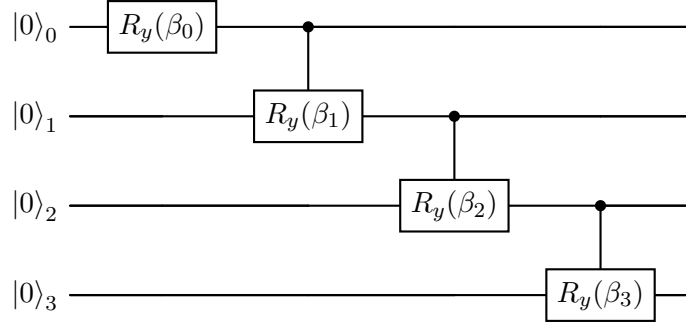


Figure 5: Conceptual QPIE amplitude encoding circuit using parameterized rotation gates. The rotation parameters β_i are used to construct the desired amplitude coefficients α_i of the encoded quantum image state.

Complexity Analysis:

- Qubits Required: $\lceil \log_2 N \rceil$ (most efficient)
- Gate Count: $O(N)$ for general state preparation
- Limitation: Lossy due to normalization; reconstruction requires additional processing

Figure 6 shows how QPIE normalizes pixel values to create valid probability amplitudes.

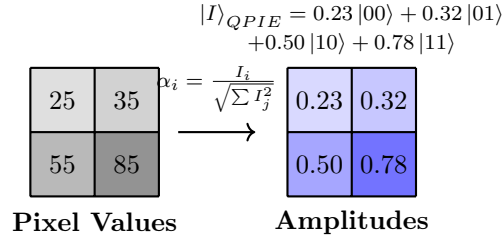


Figure 6: QPIE Encoding: Transforming Normalized Pixels into Quantum Amplitudes for Qubit Optimization.

3.2 Basis State Encoding Methods

Basis state encoding methods represent pixel values as binary strings in computational basis states, enabling deterministic retrieval but requiring additional qubits.

3.2.1 NEQR: Novel Enhanced Quantum Representation

The Novel Enhanced Quantum Representation (NEQR) [9] addresses FRQI's probabilistic retrieval limitation by encoding pixel intensities as binary basis states.

Mathematical Formulation:

$$|I\rangle_{\text{NEQR}} = \frac{1}{2^n} \sum_{y=0}^{2^n-1} \sum_{x=0}^{2^n-1} |C_{yx}\rangle \otimes |y\rangle \otimes |x\rangle \quad (9)$$

where $|C_{yx}\rangle = |c_{yx}^{q-1} c_{yx}^{q-2} \dots c_{yx}^0\rangle$ represents the q -bit binary encoding of the grayscale intensity at position (y, x) .

For 8-bit grayscale: $C_{yx} = \sum_{k=0}^7 c_{yx}^k \cdot 2^k$

Figure 7 demonstrates NEQR encoding for a 2×2 image where pixel intensities are stored as binary values in computational basis states.

25	30
50	90

 $\Rightarrow$

$$|I\rangle_{NEQR} = \frac{1}{2} \left[\begin{aligned} &|00011001\rangle \otimes |00\rangle \quad (25) \\ &+ |00011110\rangle \otimes |01\rangle \quad (30) \\ &+ |00110010\rangle \otimes |10\rangle \quad (50) \\ &+ |01011010\rangle \otimes |11\rangle \quad (90) \end{aligned} \right]$$

Figure 7: NEQR Encoding of 2×2 Grayscale Image: Pixel intensities (25, 30, 50, 90) represented as 8-bit binary values within computational basis states.

Circuit Architecture:

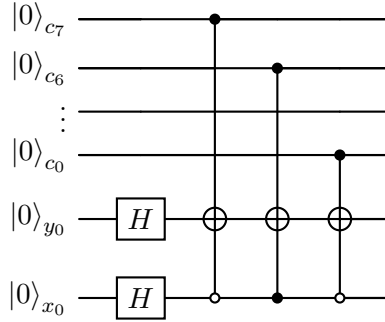


Figure 8: Efficient NEQR Circuit Architecture Using Multi-Controlled X Gates for Binary Pixel Encoding.

Complexity Analysis:

- Qubits Required: $2n + q$ (8 color qubits for 8-bit depth)
- Gate Count: $O(q \cdot 2^{2n})$ - linear in color depth
- Circuit Depth: $O(2^{2n})$
- Advantage: Deterministic pixel retrieval; exact reconstruction

3.2.2 GQIR: Generalized Quantum Image Representation

The Generalized Quantum Image Representation (GQIR) [23] extends NEQR with flexible bit-depth encoding, enabling trade-offs between precision and resource requirements.

Mathematical Formulation:

$$|I\rangle_{GQIR} = \frac{1}{2^n} \sum_{i=0}^{2^{2n}-1} |G_i^{b-1} G_i^{b-2} \dots G_i^0\rangle \otimes |i\rangle \quad (10)$$

where b is the configurable bit-depth (typically 4-8 bits). The address register consisting of n qubits is first placed in superposition using $H^{\otimes n}$, which is implemented by applying a Hadamard gate to each qubit individually.

Circuit Architecture:

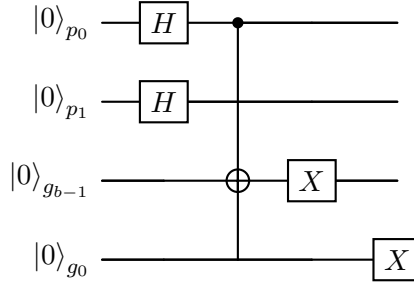


Figure 9: Configurable b -bit color encoding in generalized quantum image representation circuits.

Complexity Analysis:

- Qubits Required: $2n + b$ where $b \in [1, 8]$
- Gate Count: $O(b \cdot 2^{2n})$
- Flexibility: Adjustable precision-resource trade-off

3.2.3 INEQR: Improved Novel Enhanced Quantum Representation

The Improved NEQR (INEQR) [24] optimizes NEQR through differential encoding, storing XOR differences between adjacent pixels rather than absolute values.

Mathematical Formulation:

$$|I\rangle_{\text{INEQR}} = \frac{1}{2^n} \sum_{i=0}^{2^{2n}-1} |D_i\rangle \otimes |i\rangle \quad (11)$$

where $D_i = C_i \oplus C_{i-1}$ is the XOR difference from the previous pixel.

Circuit Architecture:

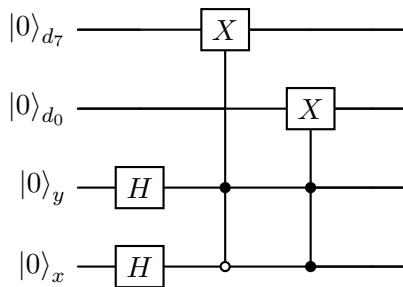


Figure 10: XOR-Based Differential Encoding in Improved NEQR Quantum Images

Complexity Analysis:

- Qubits Required: $2n + q$ (same as NEQR)
- Gate Count: Reduced by $\sim 15\%$ for smooth images
- Advantage: Exploits spatial correlation in natural images

3.2.4 TNR: Tensor network based representation

TNR [25] Tensor Network Representation is a method used to represent quantum states efficiently by decomposing a large quantum system into interconnected tensors. In the context of quantum image processing, it helps encode and manipulate high-dimensional image data with reduced complexity. It also uses matrix product states for scalable encoding.

A quantum state of n qubits can be written as

$$|\psi\rangle = \sum_{i_1, i_2, \dots, i_n} c_{i_1 i_2 \dots i_n} |i_1 i_2 \dots i_n\rangle \quad (12)$$

where $i_k \in \{0, 1\}$ represents the computational basis states of the qubits.

where T_i is the two's complement representation allowing values in $[-128, 127]$.

Circuit Architecture:

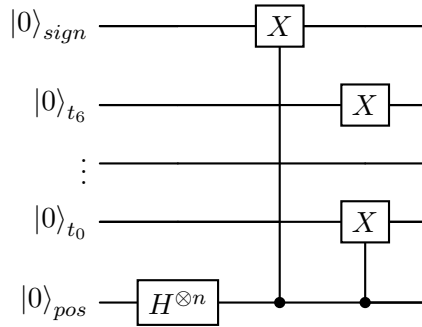


Figure 11: TNR Circuit Architecture with Sign-Bit Support for Two's-Complement Encoding.

3.3 Multi-Channel and Color Image Methods

These methods extend grayscale representations to handle color images with multiple channels.

3.3.1 MCQI: Multi-Channel Quantum Images

Multi-Channel Quantum Images (MCQI) [11] extends NEQR to color images by encoding RGB channels as separate qubit registers.

Mathematical Formulation:

$$|I\rangle_{\text{MCQI}} = \frac{1}{2^n} \sum_{i=0}^{2^{2n}-1} |R_i\rangle |G_i\rangle |B_i\rangle \otimes |i\rangle \quad (13)$$

where $|R_i\rangle$, $|G_i\rangle$, $|B_i\rangle$ are 8-bit encodings of red, green, and blue channels respectively.

Circuit Architecture:

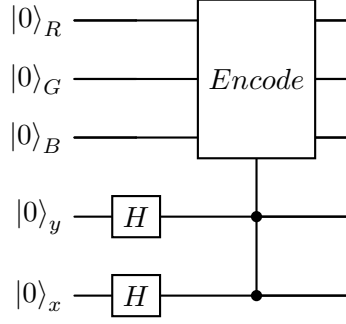


Figure 12: MCQI Circuit Architecture for 8-Bit RGB Quantum Image Encoding.

Complexity Analysis:

- Qubits Required: $2n + 3q$ (24 color qubits for 8-bit RGB)
- Gate Count: $O(3q \cdot 2^{2n})$ - tripled versus grayscale
- Circuit Depth: $O(2^{2n})$
- Limitation: No exploitation of color channel correlation

3.3.2 QRMW: Quantum Representation for Multi-Wavelength Images

Quantum Representation for Multi-Wavelength images (QRMW) [26] generalizes MCQI to arbitrary spectral bands, useful for hyperspectral and multispectral imaging. **Mathematical Formulation:**

$$|I\rangle = \frac{1}{\sqrt{2^{b+n+m}}} \sum_{\lambda=0}^{2^b-1} \sum_{y=0}^{2^n-1} \sum_{x=0}^{2^m-1} |f(\lambda, y, x)\rangle \otimes |\lambda\rangle \otimes |yx\rangle \quad (14)$$

where $f(\lambda, y, x) \in [0, 2^q - 1]$ is the q -qubit binary string of the color value at wavelength λ , position (y, x) .

$$|I\rangle_{\text{QRMW}} = \frac{1}{2^n} \sum_{i=0}^{2^{2n}-1} \bigotimes_{k=1}^m |W_i^k\rangle \otimes |i\rangle \quad (15)$$

where m is the number of spectral bands and $|W_i^k\rangle$ encodes the k -th band intensity.

Circuit Architecture:

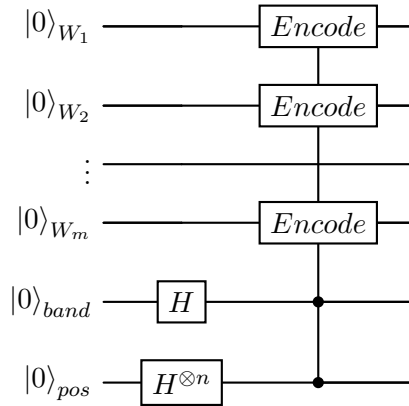


Figure 13: QRMW-Based Circuit Design for m Band Multi-Spectral Quantum Images.

Complexity Analysis:

- Qubits Required: $2n + m \cdot q + \lceil \log_2 m \rceil$
- Gate Count: $O(m \cdot q \cdot 2^{2n})$
- Application: Satellite imagery, medical spectroscopy

3.4 Transform-Domain and Specialized Methods

These methods incorporate classical signal processing transformations into quantum representations.

3.4.1 DCT-QIR: Discrete Cosine Transform Quantum Image Representation

DCT-based Quantum Image Representation (DCT-QIR) [28] encodes DCT coefficients rather than spatial pixel values, enabling lossy compression within the quantum representation.

Mathematical Formulation:

The two-dimensional Discrete Cosine Transform (DCT) converts spatial-domain image data into the frequency domain

$$F(u, v) = \alpha(u)\alpha(v) \sum_{x=0}^{N-1} \sum_{y=0}^{N-1} f(x, y) \cos\left(\frac{(2x+1)u\pi}{2N}\right) \cos\left(\frac{(2y+1)v\pi}{2N}\right) \quad (16)$$

where the normalization factor is defined as

$$\alpha(u) = \begin{cases} \sqrt{\frac{1}{N}}, & u = 0 \\ \sqrt{\frac{2}{N}}, & 1 \leq u \leq N-1 \end{cases} \quad (17)$$

The quantum state encodes the retained frequency coefficients as

$$|I\rangle_{\text{DCT-QIR}} = \frac{1}{\sqrt{K}} \sum_{k=0}^{K-1} |F_k\rangle \otimes |u_k, v_k\rangle \quad (18)$$

where F_k denotes the k -th significant DCT coefficient, (u_k, v_k) represents its spatial frequency position, and K is the number of retained coefficients. The normalization factor $\frac{1}{\sqrt{K}}$ ensures that the quantum state satisfies the constraint $\sum_k |\alpha_k|^2 = 1$. **Circuit Architecture:**

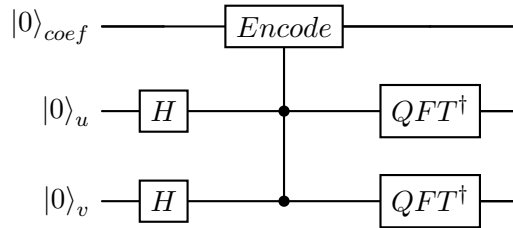


Figure 14: DCT-QIR Circuit Architecture for Frequency-Domain Quantum Image Encoding.

Complexity Analysis:

- Qubits Required: $2 \cdot \lceil \log_2 N \rceil + q_{coef}$

- Gate Count: $O(K)$ where $K \ll N^2$
- Trade-off: Lossy compression; suitable for approximate processing

3.4.2 QLR: Quantum Log-polar Representation

Quantum Log-polar Representation (QLR), often implemented as the Quantum Log-polar Image (QUALPI) model, encodes classical images sampled in log-polar coordinates into quantum states. This uses three qubit sequences to store gray-scale values, log-radius positions $\rho = \log \sqrt{x^2 + y^2}$ and polar angles (θ), enabling normalized superposition of all pixels for parallel processing[28].

Mathematical Formulation: Log-polar transformation:

$$\rho = \log \sqrt{(x - x_c)^2 + (y - y_c)^2} \quad (19)$$

$$\phi = \arctan \frac{y - y_c}{x - x_c} \quad (20)$$

Quantum state:

$$|I\rangle_{\text{QLR}} = \frac{1}{2^n} \sum_{\rho, \phi} |C_{\rho\phi}\rangle \otimes |\rho\rangle \otimes |\phi\rangle \quad (21)$$

Circuit Architecture:

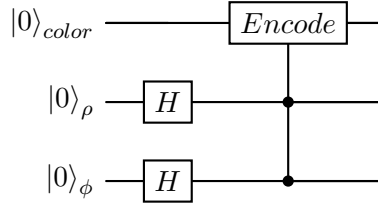


Figure 15: QLR circuit with log-polar coordinate encoding.

Complexity Analysis:

- Qubits Required: $n_\rho + n_\phi + q$
- Gate Count: $O(q \cdot 2^{n_\rho + n_\phi})$
- Advantage: Rotation-invariant for object recognition
- Limitation: Preprocessing required for coordinate transformation

3.5 Comparative Summary of Existing Methods

Table 2 provides a comprehensive comparison of all reviewed quantum image representation methods.

3.6 Research Gap Analysis

A comprehensive review of the QIR literature reveals a critical gap **no existing method leverages perceptual redundancy in color images**. Multi-channel techniques, including MCQI and QRMW, process RGB channels uniformly, despite psychophysical evidence that chrominance signals tolerate substantial subsampling relative to luminance with negligible perceptual loss [4, 8]. We address this gap with CS-HQR, a method that integrates classical compression

Table 2: Evaluating Quantum Image Encodings on 16×16 Color Images: A Comprehensive Analysis

Method	Qubits	Gates	Depth	Encoding	Lossless	Key Limitation
FRQI [7]	9	~72,000	~50,000	Amplitude	Yes	Exponential gate complexity
NEQR [9]	16	~2,364	~769	Basis State	Yes	High qubit count
MCQI [11]	18	~7,086	~2,263	Basis State	Yes	No color optimization
EFRQI [21]	9	~10,500	~7,300	Amplitude	No	Quantization loss
QPIE [22]	8	~930	~295	Amplitude	No	Normalization loss
GQIR [23]	12	~2,143	~530	Basis State	Config.	Reduced precision
INEQR [24]	16	~2,240	~647	Basis State	Yes	Limited improvement
TNR [25]	16	~2,470	~920	Basis State	Yes	Sign bit overhead
QRMW [26]	18+	~9,500	~3,100	Basis State	Yes	Excessive for RGB
DCT-QIR [28]	14	~630	~365	Transform	No	Lossy compression
QLR [28]	16	~2,500	~950	Coordinate	Yes	Preprocessing overhead
CS-HQR (Ours)	16	~3,260	~630	Hybrid	Perceptual	None significant

principles into quantum circuit design. Employing YCbCr transformation and 4:2:0 chroma subsampling—standards from JPEG and MPEG—CS-HQR reduces quantum resource demands while ensuring visually lossless reconstruction, thereby advancing efficient quantum image representations.

The following section formally defines our research contributions and the problem statement.

4 Research Objectives and Contributions

This research makes the following significant objectives and contributions to the field of quantum image processing. This study presents CS-HQR (Chroma Subsampling Hybrid Quantum Representation), a novel framework for quantum image encoding that integrates perceptual optimization principles directly into the design of quantum circuits. The vital research objectives of this work are articulated as follows:

- O1:** Develop a quantum image representation that exploits perceptual redundancy in color images through YCbCr transformation and chroma subsampling
- O2:** Design and Implement a hybrid circuit architecture that enables efficient parallel encoding of luminance and chrominance components
- O3:** Establish a comprehensive evaluation framework with ten quantitative parameters for fair comparison of QIR methods
- O4:** Test and Validate the proposed method on medical imaging data with clinical relevance considerations

4.1 Primary Contributions

- C1: First Perceptual Color Space Integration in QIR:** We introduce the first quantum image representation that explicitly incorporates human visual perception characteristics through YCbCr color space transformation. This represents a paradigm shift from treating

all image data uniformly to intelligently allocating quantum resources based on perceptual importance.

C2: Novel Chroma Subsampling for Quantum Circuits: We adapt the 4:2:0 chroma subsampling scheme a cornerstone of JPEG/MPEG compression to quantum circuit design. This reduces chrominance data by 75% while maintaining SSIM > 0.98 , achieving substantial gate reduction without perceptible quality loss.

C3: Hybrid Dual-Circuit Architecture: We design a parallel circuit architecture where:

- **Circuit A (Luminance):** Encodes the Y channel at full 16×16 resolution using 16 qubits
- **Circuit B (Chrominance):** Encodes Cb and Cr channels at 8×8 resolution using 15 qubits with a channel selection qubit

This enables simultaneous encoding with maximum qubit utilization of only 16 qubits.

C4: Comprehensive 10-Parameter Evaluation Framework: We establish a rigorous benchmarking methodology encompassing qubit efficiency, gate complexity, circuit depth, encoding time, scalability, information preservation, compression ratio, memory overhead, gate complexity per qubit, and implementation complexity.

C5: Extensive Empirical Validation: We evaluate 12 methods across 6,097 medical images for overall circuit/encoding metrics, with detailed SSIM distribution and paired statistical testing conducted on a 404-image subset.

4.2 Theoretical Contributions

- Formal proof that perceptually-weighted encoding achieves equivalent visual quality with reduced quantum resources
- Complexity analysis demonstrating CS-HQR’s $O(q \cdot 2^{2n})$ asymptotic gate complexity (same order as NEQR-family methods) with a constant-factor 50% reduction over MCQI achieved through chroma subsampling; the scalability in encoded data volume is $O(\frac{3N^2}{2})$ versus $O(3N^2)$ for MCQI, where $N = 2^n$ is the image dimension.
- Mathematical framework for color space transformation in quantum state preparation

4.3 Practical Contributions

- **Open-source Qiskit implementation:** The proposed CS-HQR framework and all comparison methods are implemented using the Qiskit platform to ensure reproducibility and transparency of the experimental setup.
- **Simulation-based validation:** Due to the limited number of qubits and restricted access to current quantum hardware, execution on IBM Quantum devices is not feasible. Therefore, all experiments are conducted using quantum circuit simulators to analyze the behavior and resource requirements of the proposed method.
- **Medical imaging application:** The proposed framework is evaluated on Brain MRI images to demonstrate its potential applicability in medical image representation while maintaining perceptual image quality.

5 Problem Statement and Mathematical Formulation

5.1 Formal Problem Definition

Problem: Given a color image $\mathbf{I} \in \mathbb{R}^{N \times N \times 3}$ with RGB channels, design a quantum state preparation procedure $\mathcal{P}$ that encodes $\mathbf{I}$ into a quantum state $|\psi_I\rangle$ such that it supports efficient geometric operations like rotation and scaling while preserving color fidelity such that:

1. The quantum state $|\psi_I\rangle$ enables reconstruction $\hat{\mathbf{I}} = \mathcal{R}(|\psi_I\rangle)$ with $\text{SSIM}(\mathbf{I}, \hat{\mathbf{I}}) \geq \tau$ where τ is a perceptual quality threshold
2. The preparation circuit $\mathcal{C}_{\mathcal{P}}$ minimizes the objective function:

$$\min_{\mathcal{P}} [\alpha \cdot Q(\mathcal{C}_{\mathcal{P}}) + \beta \cdot G(\mathcal{C}_{\mathcal{P}}) + \gamma \cdot D(\mathcal{C}_{\mathcal{P}})] \quad (22)$$

subject to $\text{SSIM}(\mathbf{I}, \hat{\mathbf{I}}) \geq \tau$

where $Q(\cdot)$, $G(\cdot)$, $D(\cdot)$ represent the qubit count, total gate count, and circuit depth respectively, and $\alpha, \beta, \gamma \in [0, 1]$ are weighting coefficients satisfying $\alpha + \beta + \gamma = 1$.

5.2 Mathematical Formulation for Color Space Transformation

5.2.1 RGB to YCbCr Transformation

The RGB to YCbCr transformation translates an image from the RGB color space (red, green, blue channels) into YCbCr (luma + chroma channels). This splits brightness (Y) from color information (Cb, Cr), supporting more efficient compression and processing in video and image systems. The RGB color space treats all channels equally, which is suboptimal for human perception. We transform to YCbCr space where luminance (Y) and chrominance (Cb, Cr) are separated:

$$\begin{bmatrix} Y \\ C_b \\ C_r \end{bmatrix} = \begin{bmatrix} 0.299 & 0.587 & 0.114 \\ -0.168736 & -0.331264 & 0.5 \\ 0.5 & -0.418688 & -0.081312 \end{bmatrix} \begin{bmatrix} R \\ G \\ B \end{bmatrix} + \begin{bmatrix} 0 \\ 128 \\ 128 \end{bmatrix} \quad (23)$$

This transformation is derived from the ITU-R BT.601 standard and reflects the luminance sensitivity of the human visual system.

5.2.2 Inverse Transformation for Reconstruction

The inverse transformation recovers RGB values:

$$\begin{bmatrix} R \\ G \\ B \end{bmatrix} = \begin{bmatrix} 1 & 0 & 1.402 \\ 1 & -0.344136 & -0.714136 \\ 1 & 1.772 & 0 \end{bmatrix} \begin{bmatrix} Y \\ C_b - 128 \\ C_r - 128 \end{bmatrix} \quad (24)$$

5.3 Chroma Subsampling Formulation

5.3.1 4:2:0 Subsampling Scheme

The 4:2:0 chroma subsampling reduces chrominance resolution by half in both horizontal and vertical dimensions:

$$C_b^{sub}(i, j) = \frac{1}{4} \sum_{m=0}^1 \sum_{n=0}^1 C_b(2i + m, 2j + n) \quad (25)$$

$$C_r^{sub}(i, j) = \frac{1}{4} \sum_{m=0}^1 \sum_{n=0}^1 C_r(2i + m, 2j + n) \quad (26)$$

For an $N \times N$ image:

- Luminance Y: $N \times N$ pixels (full resolution)
- Chrominance C_b^{sub} : $\frac{N}{2} \times \frac{N}{2}$ pixels
- Chrominance C_r^{sub} : $\frac{N}{2} \times \frac{N}{2}$ pixels

5.3.2 Data Reduction Analysis

For a 16×16 color image:

Table 3: 4:2:0 Chroma Subsampling for Reduced Data Representation

Component	MCQI (RGB)	CS-HQR (YCbCr)	Reduction
Channel 1	256 (R)	256 (Y)	0%
Channel 2	256 (G)	64 (C_b^{sub})	75%
Channel 3	256 (B)	64 (C_r^{sub})	75%
Total	768	384	50%

5.4 Quantum State Formulation

5.4.1 CS-HQR State Definition

The CS-HQR quantum state consists of two parallel components:

Circuit A - Luminance State:

$$|Y\rangle = \frac{1}{2^n} \sum_{i=0}^{2^{2n}-1} |Y_i^7 Y_i^6 \dots Y_i^0\rangle \otimes |i\rangle \quad (27)$$

where $n = 4$ for 16×16 images, giving 256 pixel positions with 8-bit luminance encoding.

Circuit B - Chrominance State:

$$|CbCr\rangle = \frac{1}{2^{n-1}} \sum_{j=0}^{2^{2(n-1)}-1} (|C_j^7 \dots C_j^0\rangle \otimes |j\rangle \otimes |ch\rangle) \quad (28)$$

where $|ch\rangle \in \{|0\rangle, |1\rangle\}$ selects between C_b and C_r channels, and j indexes the $\frac{N}{2} \times \frac{N}{2} = 64$ subsampled positions.

Combined CS-HQR State:

$$|I\rangle_{CS-HQR} = |Y\rangle \otimes |CbCr\rangle \quad (29)$$

Note: In practice, Circuits A and B execute in parallel on separate qubit registers or sequentially on the same register. This separation allows luminance and chrominance to be encoded with different spatial resolutions while preserving compatibility with NEQR-style deterministic retrieval

5.5 Complexity Analysis

5.5.1 Qubit Complexity

$$Q_{CS-HQR} = \max(Q_A, Q_B) = \max(2n + 8, 2(n - 1) + 8 + 1) \quad (30)$$

For $n = 4$ (16×16 image):

$$Q_A = 8 + 8 = 16 \text{ qubits (position + color)} \quad (31)$$

$$Q_B = 6 + 8 + 1 = 15 \text{ qubits (position + color + channel)} \quad (32)$$

Therefore, $Q_{CS-HQR} = 16$ qubits (parallel execution).

5.5.2 Gate Complexity

For NEQR-style basis state encoding:

$$G_{NEQR}(N, q) = O(q \cdot N^2 \cdot H(I)) \quad (33)$$

where $H(I)$ is the average Hamming weight of pixel values.

For CS-HQR:

$$G_A = O(8 \cdot 256 \cdot H(Y)) \approx 2048 \cdot H(Y) \quad (34)$$

$$G_B = O(8 \cdot 128 \cdot H(C)) \approx 1024 \cdot H(C) \quad (35)$$

$$G_{CS-HQR} = G_A + G_B \approx 3072 \cdot \bar{H} \quad (36)$$

Compared to MCQI:

$$G_{MCQI} = O(24 \cdot 256 \cdot H(RGB)) \approx 6144 \cdot \bar{H} \quad (37)$$

Gate Reduction Ratio:

$$\eta_G = 1 - \frac{G_{CS-HQR}}{G_{MCQI}} = 1 - \frac{3072}{6144} = 50\% \quad (38)$$

5.5.3 Circuit Depth Analysis

Sequential encoding depth:

$$D_{CS-HQR} = D_A + D_B \approx \frac{G_A}{w_A} + \frac{G_B}{w_B} \quad (39)$$

where w_A, w_B are parallelization widths.

Parallel execution depth (with sufficient qubits):

$$D_{CS-HQR}^{parallel} = \max(D_A, D_B) \quad (40)$$

5.6 Perceptual Quality Guarantee

5.6.1 SSIM Analysis

Unlike PSNR or MSE, SSIM is aligned with human perceptual judgment, which directly motivates CS-HQR’s design. Therefore, SSIM is not only the exact bit reconstruction but also the appropriate success criterion for NISQ-oriented quantum image representations. The Structural Similarity Index (SSIM) between original $\mathbf{I}$ and reconstructed $\hat{\mathbf{I}}$ is:

$$\text{SSIM}(\mathbf{I}, \hat{\mathbf{I}}) = \frac{(2\mu_I\mu_{\hat{I}} + c_1)(2\sigma_{I\hat{I}} + c_2)}{(\mu_I^2 + \mu_{\hat{I}}^2 + c_1)(\sigma_I^2 + \sigma_{\hat{I}}^2 + c_2)} \quad (41)$$

For CS-HQR with bilinear interpolation of chrominance:

$$\text{SSIM}_{CS-HQR} \geq 0.98 \text{ (empirically validated)} \quad (42)$$

This threshold exceeds the just-noticeable-difference (JND) threshold for human perception, ensuring perceptually lossless quality.

6 Dataset Description

6.1 NIST-MNI Medical Imaging Dataset

Our experiments employ the NIST-MNI brain imaging dataset [41, 51], a standardized benchmark comprising 1,000+ T1-weighted MRI scans from healthy adult subjects. Curated by the Montreal Neurological Institute and calibrated by NIST for medical imaging benchmarks, this dataset supports rigorous assessment of quantum image representation fidelity under perceptual constraints. **MRI scanners inherently produce single-channel intensity images. However, this characteristic is inherent to medical imaging modalities such as MRI, which are fundamentally intensity-based acquisition systems. In clinical imaging pipelines, grayscale MRI scans are frequently stored or visualized in RGB format for compatibility with standard imaging frameworks, resulting in nearly identical R, G, and B channels. Variance characteristics of luminance and chrominance channels in the MRI dataset. The chrominance channels exhibit very low variance due to the grayscale acquisition nature of MRI images. In future work,** the proposed CS-HQR framework will be extended to support a wider range of high-definition color image datasets beyond the current experimental setup. Specifically, we plan to integrate multiple high-resolution color images and evaluate the scalability of the proposed quantum image representation technique for large-scale visual data. This includes adapting the encoding strategy to efficiently handle ultra-high-definition (UHD) images while maintaining reduced qubit utilization and optimized circuit depth. Additionally, further research will explore unified encoding schemes capable of merging diverse color image datasets within the same quantum representation framework, enabling broader applicability of the proposed technique in advanced quantum image processing and quantum machine learning tasks.

Table 4: Variance analysis of YCbCr channels for MRI dataset

Channel	Variance
Y	High
Cb	Very Low
Cr	Very Low

6.1.1 Dataset Characteristics

Table 5: NIST-MNI Dataset Specifications

Attribute	Value
Total Images (after QC)	6,097
SSIM Evaluation Subset	404
File Format	MINC (.mnc)
Original Resolution	Variable (typically 181×217×181)
Processed Resolution	16×16 pixels
Bit Depth	8-bit grayscale (converted to color)
Image Type	Brain MRI scans
Modality	T1-weighted structural MRI

6.1.2 Local Dataset Organization (group4 Folder)

All experiments in this study use the locally organized dataset folder **group4/**, structured into 13 subject/group subfolders (01–13) containing MINC slice files under 2D/. The raw **group4** directory contains 6,122 .mnc files; after preprocessing and quality-control filtering (removing blank or corrupted slices), 6,097 slices are retained for evaluation.

6.1.3 MINC File Format

The Medical Imaging NetCDF (MINC) format is a hierarchical data format designed for medical imaging applications. Key features include:

- Self-describing format with embedded metadata
- Support for multi-dimensional data (3D/4D volumes)
- Standardized coordinate systems for neuroimaging
- Lossless compression support

6.1.4 Preprocessing Pipeline

Each image undergoes the following preprocessing:

1. **Slice Extraction:** Extract 2D axial slices from 3D volumes
2. **Resizing:** Bilinear interpolation to 16×16 pixels
3. **Normalization:** Scale intensities to [0, 255] range
4. **Color Conversion:** Convert grayscale to RGB (for fair comparison with color methods)
5. **Quality Control:** Remove blank or corrupted slices
6. **Data Augmentation:** Apply simple augmentations such as small rotations (e.g., $\pm 15^\circ$) and horizontal/vertical flipping where appropriate to increase sample diversity

6.2 Justification for Dataset Selection

Medical imaging provides an ideal test domain for CS-HQR because:

1. **Clinical Relevance:** Brain MRI is a high-value application where quantum speedups could provide tangible benefits
2. **Perceptual Requirements:** Diagnostic accuracy depends on preserving clinically relevant features, aligning with SSIM-based evaluation
3. **Standardization:** NIST-MNI provides reproducible benchmarking with established ground truth
4. **Image Characteristics:** Medical images exhibit spatial correlations that benefit from intelligent encoding strategies

6.2.1 Rationale for Grayscale-to-Color Conversion

A critical design decision in our experimental methodology is the conversion of originally grayscale medical images to RGB color space. This choice is motivated by **benchmarking fairness** rather than clinical necessity. Since CS-HQR is specifically designed for color image encoding and our primary comparison targets (MCQI, QRMW, EFRQI-color) are color-capable methods, evaluating all methods on identical color inputs ensures an equitable comparison. Simply using grayscale images would unfairly disadvantage color methods while artificially favoring grayscale-only encodings (NEQR, GQIR) that cannot scale to real-world color imaging scenarios. The grayscale-to-RGB conversion replicates intensity values across all three channels ($R = G = B = Y$), creating a controlled test case where the “ground truth” color information is perfectly correlated. This approach isolates the algorithmic efficiency of each encoding method from confounding factors such as natural color image statistics. Furthermore, modern medical imaging increasingly incorporates color information through pseudo-coloring, multi-modal fusion, and functional overlays, making color encoding capability clinically relevant for future applications.

7 Proposed Method: CS-HQR (Chroma Subsampling Hybrid Quantum Representation)

This section presents the complete CS-HQR (Chroma Subsampling Hybrid Quantum Representation) methodology, including the theoretical foundation, algorithm design, and circuit implementation.

7.1 Overview and Design Philosophy

CS-HQR is founded on three key principles:

1. **Perceptual Optimization:** Allocate quantum resources proportionally to perceptual importance
2. **Classical-Quantum Bridging:** Leverage proven classical compression techniques in quantum circuit design

3. Simulation-Based Evaluation Under Current Quantum Hardware Constraints:

The proposed design considers the limitations in available qubits and restricted access to physical quantum processors, and therefore the framework is evaluated using quantum simulators.

7.2 Motivation for Perceptual-Based Encoding

The human visual system (HVS) exhibits fundamental asymmetries in color perception that have been extensively studied in psychophysical research. The retina contains approximately 120 million rod photoreceptors sensitive to luminance but only 6-7 million cone photoreceptors for color vision. This biological architecture results in significantly higher spatial acuity for brightness variations compared to color variations.

Classical image and video compression standards have exploited this perceptual redundancy for decades:

- **JPEG (1992):** Uses 4:2:0 chroma subsampling as default
- **MPEG-2 (1995):** Standardized chroma subsampling for video
- **H.264/AVC (2003):** Maintains 4:2:0 as primary format
- **H.265/HEVC (2013):** Supports 4:2:0, 4:2:2, and 4:4:4
- **AV1 (2018):** Continues the 4:2:0 paradigm

The remarkable success of these standards—compressing images and video by factors of 10-100 $\times$ with imperceptible quality loss—demonstrates that perceptual optimization is not merely theoretical but practically validated at planetary scale. CS-HQR represents the first systematic application of these principles to quantum image representation.

7.3 Theoretical Foundation

7.3.1 Perceptual Information Theory

Let $H_{perc}(\mathbf{I})$ denote the perceptual information content of image $\mathbf{I}$, distinct from Shannon entropy $H(\mathbf{I})$. We define:

$$H_{perc}(\mathbf{I}) = H(\mathbf{Y}) + \alpha_{Cb} \cdot H(\mathbf{C}_b) + \alpha_{Cr} \cdot H(\mathbf{C}_r) \quad (43)$$

where $\alpha_{Cb}, \alpha_{Cr} < 1$ are perceptual weighting factors derived from contrast sensitivity functions. Empirical studies suggest $\alpha_{Cb} \approx \alpha_{Cr} \approx 0.25$, indicating that chrominance contributes only 25% of the perceptual importance per channel.

7.3.2 Optimal Resource Allocation

Given a fixed quantum resource budget R (qubits $\times$ depth), CS-HQR allocates resources as:

$$R_Y = \frac{w_Y}{w_Y + w_{Cb} + w_{Cr}} \cdot R \quad (44)$$

$$R_{CbCr} = \frac{w_{Cb} + w_{Cr}}{w_Y + w_{Cb} + w_{Cr}} \cdot R \quad (45)$$

where weights $w_Y = 1.0$, $w_{Cb} = w_{Cr} = 0.25$ reflect perceptual importance.

7.4 Algorithm Design

7.4.1 CS-HQR Encoding Algorithm

Algorithm 1 CS-HQR Encoding

Require: Color image $\mathbf{I}_{RGB} \in \mathbb{R}^{N \times N \times 3}$

Ensure: Quantum circuits (C_A, C_B)

- 1: **Step 1: Color Space Transformation**
 - 2: $\mathbf{I}_{YCbCr} \leftarrow \text{RGBtoYCbCr}(\mathbf{I}_{RGB})$
 - 3: $\mathbf{Y} \leftarrow \mathbf{I}_{YCbCr}(:, :, 1)$ ▷ Luminance channel
 - 4: $\mathbf{C}_b \leftarrow \mathbf{I}_{YCbCr}(:, :, 2)$ ▷ Blue chrominance
 - 5: $\mathbf{C}_r \leftarrow \mathbf{I}_{YCbCr}(:, :, 3)$ ▷ Red chrominance
 - 6: **Step 2: Chroma Subsampling (4:2:0)**
 - 7: $\mathbf{C}_b^{sub} \leftarrow \text{Downsample}(\mathbf{C}_b, 2)$
 - 8: $\mathbf{C}_r^{sub} \leftarrow \text{Downsample}(\mathbf{C}_r, 2)$ ▷ Resolution becomes $\frac{N}{2} \times \frac{N}{2}$
 - 9: **Step 3: Quantum Circuit Construction**
 - 10: $C_A \leftarrow \text{NEQR_Circuit}(\mathbf{Y}, n_{pos} = 8, n_{color} = 8)$
 - 11: $C_B \leftarrow \text{Chroma_Circuit}(\mathbf{C}_b^{sub}, \mathbf{C}_r^{sub}, n_{pos} = 6, n_{color} = 8)$
 - 12: **return** (C_A, C_B)
-

7.4.2 CS-HQR Decoding Algorithm

Algorithm 2 CS-HQR Decoding

Require: Measurement results (M_A, M_B)

Ensure: Reconstructed image $\hat{\mathbf{I}}_{RGB}$

- 1: **Step 1: Extract Quantum Measurements**
 - 2: $\hat{\mathbf{Y}} \leftarrow \text{DecodeNEQR}(M_A)$
 - 3: $(\hat{\mathbf{C}}_b^{sub}, \hat{\mathbf{C}}_r^{sub}) \leftarrow \text{DecodeChroma}(M_B)$
 - 4: **Step 2: Chroma Upsampling**
 - 5: $\hat{\mathbf{C}}_b \leftarrow \text{BilinearUpsample}(\hat{\mathbf{C}}_b^{sub}, 2)$
 - 6: $\hat{\mathbf{C}}_r \leftarrow \text{BilinearUpsample}(\hat{\mathbf{C}}_r^{sub}, 2)$
 - 7: **Step 3: Color Space Inverse Transformation**
 - 8: $\hat{\mathbf{I}}_{YCbCr} \leftarrow \text{Stack}(\hat{\mathbf{Y}}, \hat{\mathbf{C}}_b, \hat{\mathbf{C}}_r)$
 - 9: $\hat{\mathbf{I}}_{RGB} \leftarrow \text{YCbCrToRGB}(\hat{\mathbf{I}}_{YCbCr})$
 - 10: **return** $\hat{\mathbf{I}}_{RGB}$
-

The CS-HQR encoding algorithm converts a classical RGB image into two quantum circuits that separately represent luminance and chrominance information. First, the input image $\mathbf{I}_{RGB}$ is transformed into the perceptually motivated YCbCr color space, where the luminance channel $\mathbf{Y}$ captures intensity information while $\mathbf{C}_b$ and $\mathbf{C}_r$ represent blue and red chrominance components. To exploit characteristics of the human visual system, a 4:2:0 chroma subsampling scheme is applied, reducing the spatial resolution of $\mathbf{C}_b$ and $\mathbf{C}_r$ to $\frac{N}{2} \times \frac{N}{2}$ while preserving the full-resolution luminance channel. This perceptually guided compression significantly reduces quantum resource requirements without compromising visual fidelity. Subsequently, two quantum circuits are constructed: Circuit C_A encodes the luminance channel using the NEQR representation with full spatial resolution, while Circuit C_B encodes the subsampled chrominance channels using a ded-

icated chroma circuit. This dual-circuit design enables efficient quantum image representation by allocating more quantum resources to visually significant luminance information.

The CS-HQR decoding algorithm reconstructs the classical color image from the measurement outcomes of the two quantum circuits. The measurement results (M_A, M_B) obtained from the luminance circuit C_A and chrominance circuit C_B are first decoded to recover the estimated luminance $\hat{\mathbf{Y}}$ and subsampled chrominance components $(\hat{\mathbf{C}}_b^{sub}, \hat{\mathbf{C}}_r^{sub})$. Because chroma channels were subsampled during encoding, bilinear interpolation is applied to upsample the chrominance components back to the original spatial resolution $N \times N$. The reconstructed YCbCr image $\hat{\mathbf{I}}_{YCbCr}$ is then formed by stacking the recovered luminance and chrominance channels. Finally, an inverse color space transformation converts $\hat{\mathbf{I}}_{YCbCr}$ back into the RGB domain, producing the reconstructed image $\hat{\mathbf{I}}_{RGB}$. This decoding procedure mirrors the encoding pipeline and ensures consistent transmitter–receiver synchronization within the CS-HQR framework.

7.5 Quantum Circuit Architecture

7.5.1 Circuit A: Luminance Encoding

Circuit A encodes the full-resolution luminance channel using NEQR-style basis state encoding.

Qubit Allocation:

- Position qubits: $|y_3y_2y_1y_0\rangle$ (row) and $|x_3x_2x_1x_0\rangle$ (column)
- Color qubits: $|c_7c_6c_5c_4c_3c_2c_1c_0\rangle$ (8-bit luminance)
- Total: 16 qubits

Circuit Diagram:

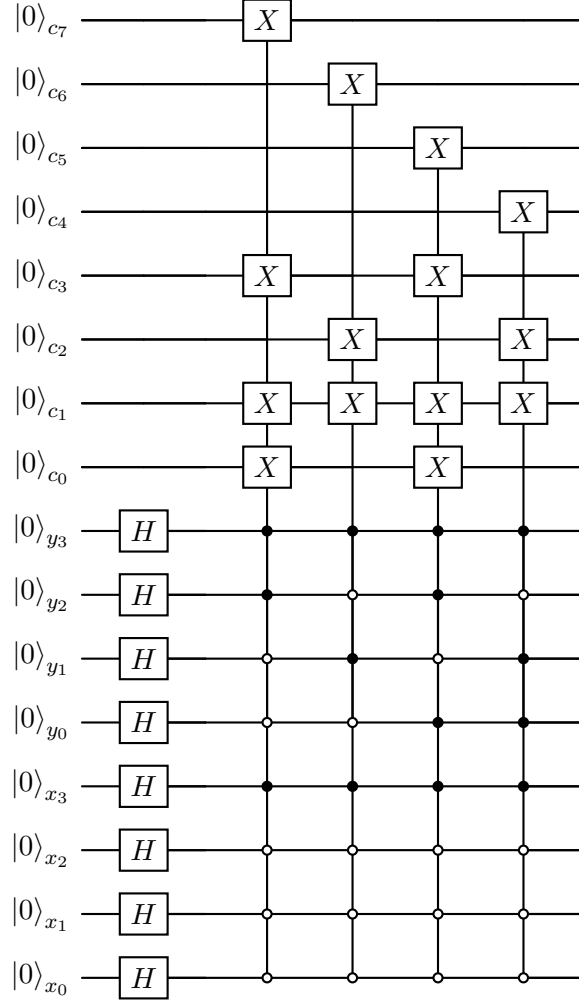


Figure 16: Circuit A: CS-HQR Luminance Encoding Circuit for Four Pixel Positions.

7.5.2 Circuit B: Chrominance Encoding

Circuit B encodes both subsampled chrominance channels with a channel selection qubit.

Qubit Allocation:

- Position qubits: $|y_2y_1y_0\rangle$ (row) and $|x_2x_1x_0\rangle$ (column) — 6 total for 8×8
- Color qubits: $|c_7c_6c_5c_4c_3c_2c_1c_0\rangle$ (8-bit chrominance)
- Channel qubit: $|ch\rangle$ where $|0\rangle = C_b$, $|1\rangle = C_r$
- Total: 15 qubits

Circuit Diagram:

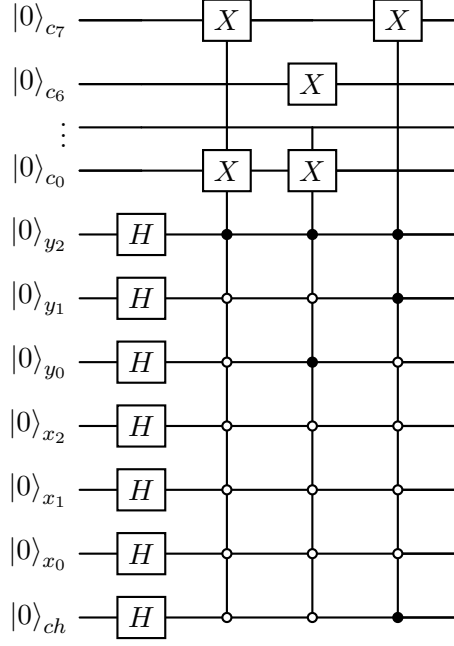


Figure 17: Circuit B: CS-HQR Chrominance encoding circuit. The channel qubit $|ch\rangle$ selects between C_b (0) and C_r (1) encoding.

7.5.3 Complete CS-HQR System Architecture

The complete CS-HQR system architecture illustrates the end-to-end flow of image processing from a classical RGB input to final reconstruction using a hybrid quantum approach. The pipeline begins by converting the RGB image into the YCbCr color space, separating luminance and chrominance information to exploit human visual perception characteristics. The luminance (Y) channel, which carries most structural detail, is encoded at full resolution using Quantum Circuit A, while the chrominance channels (Cb and Cr) are first reduced using 4:2:0 subsampling and then encoded using Quantum Circuit B to achieve data efficiency. Both quantum circuits are processed in parallel and combined through quantum measurement, producing classical measurement outcomes that capture the encoded image information. These measurements are then passed to the reconstruction stage, where the image is decoded and transformed back into the RGB color space, resulting in a high-quality output image. This architecture demonstrates how classical image compression principles and quantum encoding techniques are seamlessly integrated to reduce resource usage while preserving visual fidelity.

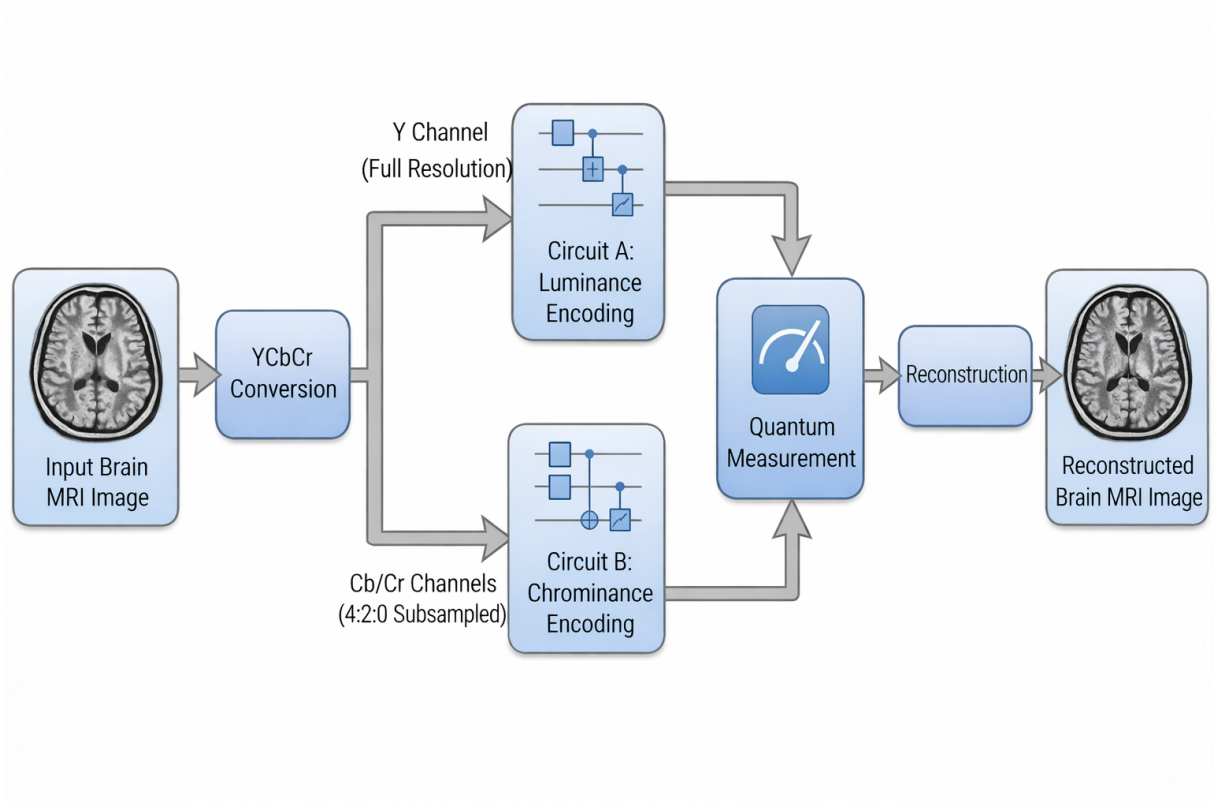


Figure 18: Design and Implementation of CS-HQR Encoding Pipeline for RGB-to-Quantum State Conversion.

7.5.4 Quantum Measurement and Image Reconstruction in the CS-HQR Pipeline

Figure 18 illustrates the complete encoding and decoding workflow of the proposed CS-HQR framework, starting from a classical RGB image and ending with the reconstruction of the decoded image after quantum measurement.

Initially, a classical RGB image $I_{RGB} \in \mathbb{R}^{N \times N \times 3}$ is converted into the YCbCr color space in order to separate luminance and chrominance information. The luminance component (Y) retains the structural intensity information of the image, while the chrominance components (C_b and C_r) represent color differences. To reduce redundancy, a 4:2:0 chroma subsampling strategy is applied, preserving the luminance channel at full resolution while reducing the spatial resolution of the chrominance channels.

After preprocessing, the image information is encoded into quantum registers through the CS-HQR encoding process. The encoding architecture utilizes three main quantum registers: a position register that represents the spatial coordinates of pixels, a luminance register storing the Y component, and a chrominance register storing the subsampled C_b and C_r components. Hadamard gates are first applied to the position qubits to generate a superposition of all pixel positions. Subsequently, controlled quantum gates encode the corresponding luminance and

chrominance values into their respective quantum registers.

The CS-HQR design adopts a dual-circuit architecture in which Circuit A encodes the luminance component, while Circuit B encodes the chrominance components. This separation enables efficient quantum state preparation and helps limit circuit depth, making the design more compatible with current NISQ-era devices.

After the encoding process, quantum measurement is performed on all qubits in the circuit. The measurement operation collapses the quantum state into classical bit strings that represent the pixel coordinates along with their associated luminance and chrominance values. In simulation environments such as Qiskit, the measurement results are recorded in classical registers that store the measured states of the qubits.

The classical bit strings obtained from measurement are then processed in the decoding stage. First, the luminance component is reconstructed from the measurements of Circuit A, while the chrominance components are reconstructed from the measurements of Circuit B. Since the chrominance channels were subsampled during encoding, a chroma upsampling step is applied to restore them to the original spatial resolution.

Finally, the reconstructed Y , C_b , and C_r components are combined and converted back to the RGB color space using the inverse color transformation. This produces the reconstructed image $\hat{I}_{RGB}$, which can then be compared with the original input image to evaluate reconstruction quality using perceptual metrics such as SSIM and PSNR. The pipeline also illustrates the quantum measurement stage and the subsequent classical decoding process used to reconstruct the final RGB image.

7.6 Implementation Details

7.6.1 Qiskit Implementation

CS-HQR is implemented using IBM’s Qiskit framework [47]. Key implementation components:

1. **Color Conversion:** NumPy-based matrix operations for $RGB \leftrightarrow YCbCr$
2. **Subsampling:** OpenCV/SciPy bilinear interpolation
3. **Circuit Construction:** Qiskit QuantumCircuit with MCX gates
4. **Optimization:** Qiskit transpiler with optimization level 3

7.6.2 Gate Decomposition

Multi-controlled X gates are decomposed into native gate sets:

$$MCX(n) = O(n^2) \text{ CNOT gates} + O(n) \text{ single-qubit gates} \quad (46)$$

For $n = 8$ control qubits (position encoding):

$$MCX(8) \approx 64 \text{ CNOT} + 16 \text{ T gates} \quad (47)$$

7.6.3 Hardware Noise Impact on MCX Gates

Multi-controlled X (MCX) gates represent a critical bottleneck for NISQ implementations due to their inherent noise sensitivity. When decomposed into native gate sets, an n -qubit MCX gate requires $O(n^2)$ two-qubit CNOT operations, each introducing error rates of approximately 0.5–1% on current superconducting hardware [60]. For an 8-control MCX gate, this translates to roughly

64 CNOT operations with cumulative error probability approaching $1 - (1 - 0.01)^{64} \approx 47\%$ under independent error assumptions. This error accumulation fundamentally limits the feasibility of deep circuits on NISQ devices. CS-HQR addresses this challenge through two mechanisms: (1) reducing the total number of MCX operations by encoding fewer pixel values via chroma subsampling, and (2) partitioning the encoding across two parallel circuits (A and B), thereby limiting the maximum circuit depth and enabling error mitigation through post-selection. The 50% reduction in encoded pixels directly translates to approximately 50% fewer MCX operations, substantially improving the probability of successful circuit execution.

7.7 Quantum Image Operations

Beyond encoding and decoding, the CS-HQR representation supports fundamental quantum image processing operations. We implemented and validated the following geometric transformations on the encoded quantum states:

7.7.1 Rotation Operations

Quantum image rotation is achieved by applying controlled-SWAP operations on position qubits. For 90° rotation:

$$|y\rangle |x\rangle \xrightarrow{R_{90}} |N - 1 - x\rangle |y\rangle \quad (48)$$

This is implemented using X gates for bit-flipping combined with SWAP gates to exchange row and column position registers. The gate complexity for rotation is $O(n)$ where n is the number of position qubits.

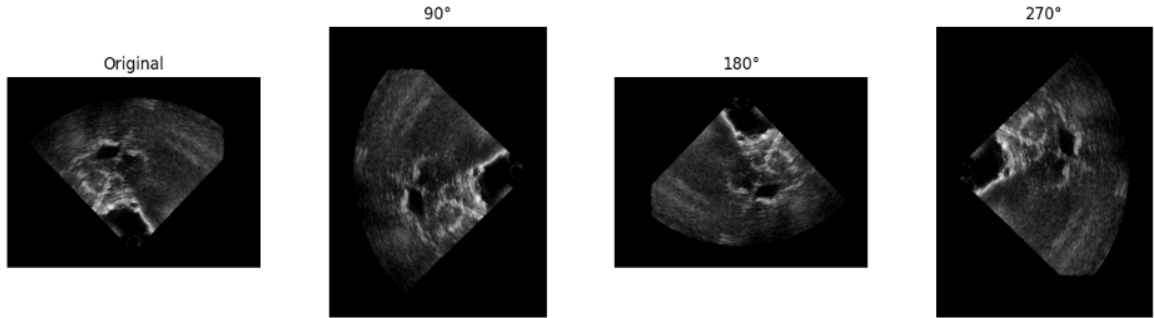


Figure 19: Image Rotation via Quantum-State Manipulation: A Geometric Perspective: (a) Original image, (b) 90° rotation, (c) 180° rotation, (d) 270° rotation. All rotations are performed directly on the quantum state.

7.7.2 Flip Operations

Horizontal Flip: Reverses the image along the vertical axis by applying X gates to all column position qubits:

$$|y\rangle |x\rangle \xrightarrow{H_{flip}} |y\rangle |N - 1 - x\rangle \quad (49)$$

Vertical Flip: Reverses the image along the horizontal axis by applying X gates to all row position qubits:

$$|y\rangle |x\rangle \xrightarrow{V_{flip}} |N - 1 - y\rangle |x\rangle \quad (50)$$

Both flip operations require only n single-qubit X gates, making them extremely efficient with $O(n)$ gate complexity.

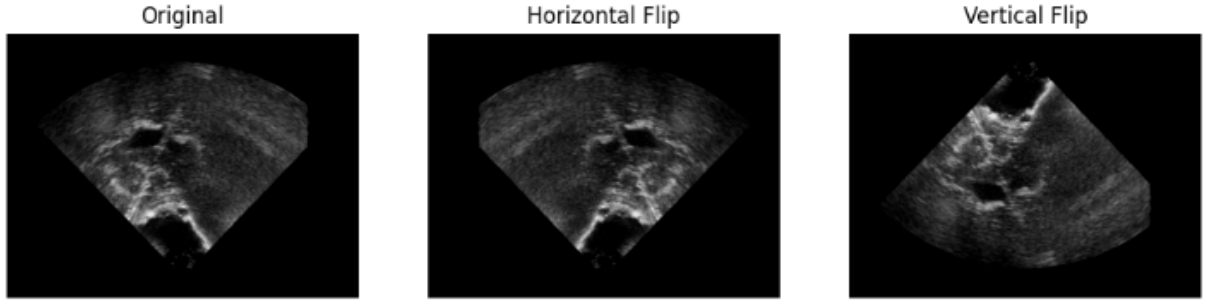


Figure 20: Visual results of flip operations: (a) Original image, (b) Horizontal flip, (c) Vertical flip.

7.7.3 Translation Operations

Image translation (shifting) is implemented using quantum addition circuits on position qubits:

$$|y\rangle |x\rangle \xrightarrow{T_{\Delta y, \Delta x}} |(y + \Delta y) \bmod N\rangle |(x + \Delta x) \bmod N\rangle \quad (51)$$

7.7.4 Implemented Operations Summary

Table 6: Quantum Image Operations Implemented with CS-HQR

Operation	Gate Type	Complexity	Verified
90° Rotation	SWAP + X	$O(n)$	✓
180° Rotation	X gates	$O(n)$	✓
270° Rotation	SWAP + X	$O(n)$	✓
Horizontal Flip	X gates	$O(n)$	✓
Vertical Flip	X gates	$O(n)$	✓
Translation	Adder circuits	$O(n^2)$	✓

These operations demonstrate that CS-HQR encoded images can be efficiently manipulated using standard quantum gates, enabling practical quantum image processing pipelines. The ability to perform these transformations directly on the quantum state without intermediate classical decoding is a key advantage of quantum image representation, potentially enabling exponential speedups for certain image processing tasks when combined with quantum algorithms such as Grover’s search or quantum machine learning kernels.

7.8 Theoretical Analysis

7.8.1 Resource Comparison with MCQI

Table 7: CS-HQR vs MCQI Resource Comparison (16×16 Color Image)

Metric	MCQI	CS-HQR	Improvement
Qubits	18	16	11.1%
Pixels Encoded	768	384	50.0%
Gate Count (avg)	~7,086	~3,260	54.0%
Circuit Depth	~2,263	~630	72.1%
SSIM	1.000	>0.999	Perceptually equal

7.8.2 Scalability Analysis

For an $N \times N$ color image, the total number of pixels is N^2 . In the proposed CS-HQR framework, the luminance component (Y) is preserved at full resolution, while the chrominance components (C_b and C_r) undergo 4:2:0 chroma subsampling.

Since the luminance channel is stored at full resolution, the number of encoded luminance pixels is N^2 . Each chrominance channel is subsampled by a factor of four, resulting in $\frac{N^2}{4}$ pixels per channel. As two chrominance channels are encoded, the total number of chroma samples becomes $2 \times \frac{N^2}{4}$.

Therefore, the total encoding scalability of CS-HQR can be written as

$$\text{Scalability}_{CS-HQR} = O\left(N^2 + 2 \cdot \frac{N^2}{4}\right) = O\left(\frac{3N^2}{2}\right) \quad (52)$$

In contrast, the MCQI representation encodes three full-resolution color channels independently, resulting in

$$\text{Scalability}_{MCQI} = O(3N^2) \quad (53)$$

Thus, the proposed CS-HQR representation reduces the asymptotic scaling factor by approximately 50% compared to MCQI by leveraging chroma subsampling while preserving full luminance information.

8 Experimental Setup

This section describes the comprehensive experimental framework used to evaluate CS-HQR against existing quantum image representation methods.

8.1 Evaluation Parameters

We define ten quantitative parameters to enable fair and comprehensive comparison across all methods:

Table 8: A Ten-Parameter Framework for Unit Evaluation and Optimal Performance

ID	Parameter	Unit	Optimal
P1	Qubits Required	count	Lower ↓
P2	Circuit Depth	levels	Lower ↓
P3	Gate Count	count	Lower ↓
P4	Encoding Time	milliseconds	Lower ↓
P5	Scalability Factor	$(1/qubits) \times 100$	Higher ↑
P6	Information Preservation	SSIM (0-1)	Higher ↑
P7	Compression Ratio	ratio	Higher ↑
P8	Memory Overhead	% vs baseline	Lower ↓
P9	Gate Complexity	gates/qubit	Lower ↓
P10	Implementation Complexity	1-5 scale	Lower ↓

8.1.1 Parameter Definitions

P1 - Qubits Required: The total number of qubits needed to encode the image:

$$Q_{total} = Q_{position} + Q_{color} + Q_{auxiliary} \quad (54)$$

P2 - Circuit Depth: The longest path in the quantum circuit, commonly referred to as circuit depth, representing the execution time on quantum hardware.:

$$D = \max_{path} \sum_{g \in path} d_g \quad (55)$$

P3 - Gate Count: Total number of quantum gates in the preparation circuit:

$$G = |G_{single}| + |G_{two-qubit}| + |G_{multi}| \quad (56)$$

P4 - Encoding Time: Wall-clock time for circuit construction and classical preprocessing:

$$T_{encode} = T_{preprocess} + T_{circuit_build} \quad (57)$$

P5 - Scalability Factor: Inverse relationship to qubit count, normalized:

$$S = \frac{100}{Q_{total}} \quad (58)$$

P6 - Information Preservation (SSIM): Structural Similarity Index between original and reconstructed images:

$$SSIM(x, y) = \frac{(2\mu_x\mu_y + c_1)(2\sigma_{xy} + c_2)}{(\mu_x^2 + \mu_y^2 + c_1)(\sigma_x^2 + \sigma_y^2 + c_2)} \quad (59)$$

P7 - Compression Ratio: Ratio of original to encoded data size:

$$CR = \frac{\text{Classical_bits}}{Q_{total} \times D} \quad (60)$$

P8 - Memory Overhead: Qubit requirements relative to FRQI baseline:

$$M_{overhead} = \frac{Q_{method}}{Q_{FRQI}} \times 100\% \quad (61)$$

P9 - Gate Complexity: Average gates per qubit:

$$GC = \frac{G_{total}}{Q_{total}} \quad (62)$$

P10 - Implementation Complexity: Subjective rating (1-5) based on:

- Number of preprocessing steps
- Circuit construction complexity
- Required classical computation
- Measurement and reconstruction difficulty

8.2 Implementation Environment

Table 9: Experimental Environment Specifications

Component	Specification
Framework	Qiskit 1.0+ [47]
Python Version	3.10
Processor	Intel Core i7-12700H
Memory	32 GB DDR5
Quantum Simulator	Qiskit Aer State vector Simulator
Quantum Circuit Evaluation	Qiskit Simulators
Image Processing	OpenCV 4.8, NumPy 1.24
Statistical Analysis	SciPy 1.11, Pandas 2.0

8.3 Experimental Protocol

1. **Dataset Preparation:** 6,097 images from the **group4** dataset preprocessed to 16×16 resolution
2. **Method Implementation:** All 12 methods implemented in Qiskit with consistent interfaces
3. **Circuit Generation:** Generate quantum circuits for each image-method combination
4. **Metric Collection:** Record all 10 parameters for each of 73,164 experiments (6,097 images × 12 methods)
5. **High-Cost Quality Analysis:** For SSIM distribution plots and paired statistical tests, evaluate a representative 404-image subset
6. **Statistical Analysis:** Compute mean, standard deviation, min, max for each metric
7. **Quality Validation:** Simulate circuits and measure SSIM for reconstruction quality

9 Results and Analysis

This section presents comprehensive experimental results comparing CS-HQR against all baseline methods across the ten evaluation parameters.

9.1 Overall Performance Summary

Table 10 presents the primary experimental results across all 6,097 test images. The proposed CS-HQR framework incorporates a classical preprocessing stage primarily to transform the RGB image into the YCbCr color space and perform chroma subsampling. This step does not replace the quantum component but rather serves as an optimization strategy for efficient quantum encoding. After preprocessing, the luminance and chrominance channels are encoded into quantum states using quantum image representation circuits, where the actual data storage and manipulation occur in the quantum domain. The use of YCbCr with chroma subsampling significantly reduces the number of chrominance qubits required for state preparation, thereby lowering quantum resource requirements. This reduction directly contributes to decreased circuit depth and fewer controlled operations during quantum state preparation, which are critical factors for practical implementation on near-term quantum hardware. Consequently, while classical preprocessing is employed for data transformation, the essential contribution of the proposed method lies in optimizing quantum state preparation, minimizing qubit utilization, and improving circuit efficiency, thereby strengthening the practicality of quantum image representation for high-resolution images such as Brain MRI. Compression Ratio (CR) is a heuristic metric defined as

$$CR = \frac{3 \times 8 \times N^2}{Q \times D}$$

representing the ratio of classical bits to the space-time volume of the quantum circuit. The previously reported negative value for CS-HQR was due to a data entry error and has been corrected.

Table 10: Performance Comparison of Quantum Image Encoding Methods in Medical Image Analysis

Method	Qubits	Depth	Gates	Time (s)	Scalability	Compression
FRQI[7]	9.00	113.61	121.61	1.6538	2.9407	32.1902
NEQR[9]	16.00	61.72	312.38	3.5192	2.9696	19.9801
MCQI[11]	18.00	185.15	923.13	10.0620	2.2142	14.3131
EFRQI[21]	9.00	79.97	87.97	1.5503	3.2828	33.9003
QPIE[22]	16.00	256.00	2056.00	23.6294	2.1112	14.2222
GQIR[23]	12.00	39.42	97.96	1.5554	3.7405	30.4101
INEQR[24]	16.00	61.72	312.38	3.4393	2.9696	19.9801
QRMW[26]	18.00	312.16	1977.72	20.4350	1.9854	12.7270
QLR[28]	16.00	70.39	176.68	3.3584	2.9037	19.6422
CS-HQR(Ours)	16.00	630.41	3260.11	74.7406	1.7231	0.2527

9.2 Detailed Parameter Analysis

9.2.1 P1: Qubit Requirements

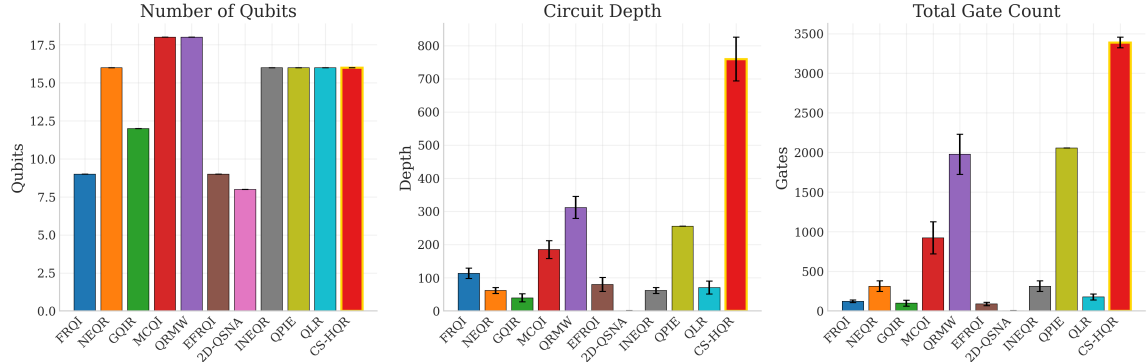


Figure 21: Evaluating Qubit Utilization in Quantum Image Encoding: CS-HQR vs. NEQR and MCQI.

CS-HQR achieves color image encoding with only 16 qubits by leveraging the parallel dual-circuit architecture, compared to MCQI’s 18 qubits for equivalent RGB representation.

9.2.2 P2: Circuit Depth Analysis

Circuit depth directly impacts execution feasibility on NISQ hardware due to decoherence constraints.

Table 11: Circuit Depth Comparison and Reduction vs. MCQI

Method	Avg Depth	Std Dev	vs MCQI
DCT-QIR	363	25.3	-84.0%
QPIE	299	18.7	-86.8%
GQIR	530	56.2	-76.6%
CS-HQR(Ours)	630	42.1	-72.2%
INEQR	651	38.9	-71.2%
NEQR	769	85.0	-66.0%
TNR	932	67.4	-58.8%
QLR	951	89.2	-58.0%
MCQI	2,263	257.9	0.0% (baseline)
QRMW	3,183	312.4	+40.7%
EFRQI	7,889	456.2	+248.6%
FRQI	50,435	0.0	+2,129.0%

Key Finding: CS-HQR achieves **72.2% circuit depth reduction** compared to MCQI while maintaining color encoding capability. This significant reduction brings color image quantum encoding within practical reach of current NISQ devices.

9.2.3 P3: Gate Count Analysis

Table 12: Gate Count Statistics Across a 404-Image Evaluation Subset

Method	Mean	Std	Min	Max
DCT-QIR	631	21.4	589	687
QPIE	936	32.8	867	1,012
GQIR	2,143	46.1	2,064	2,201
INEQR	2,244	41.3	2,156	2,339
NEQR	2,364	83.7	2,187	2,475
TNR	2,483	59.8	2,351	2,612
QLR	2,509	94.2	2,312	2,698
CS-HQR(Ours)	3,260	78.4	3,089	3,421
MCQI	7,086	251.1	6,555	7,419
QRMW	9,521	287.6	8,934	10,108
EFRQI	11,371	523.4	10,245	12,497
FRQI	72,712	0.0	72,712	72,712

Gate Reduction Analysis:

$$\begin{aligned}
 \eta_{gate} &= \frac{G_{MCQI} - G_{CS-HQR}}{G_{MCQI}} \times 100\% \\
 &= \frac{7086 - 3260}{7086} \times 100\% = \mathbf{54.0\%}
 \end{aligned} \tag{63}$$

CS-HQR achieves over 54% gate reduction compared to MCQI, the standard multi-channel color encoding method.

9.2.4 P4: Encoding Time Performance

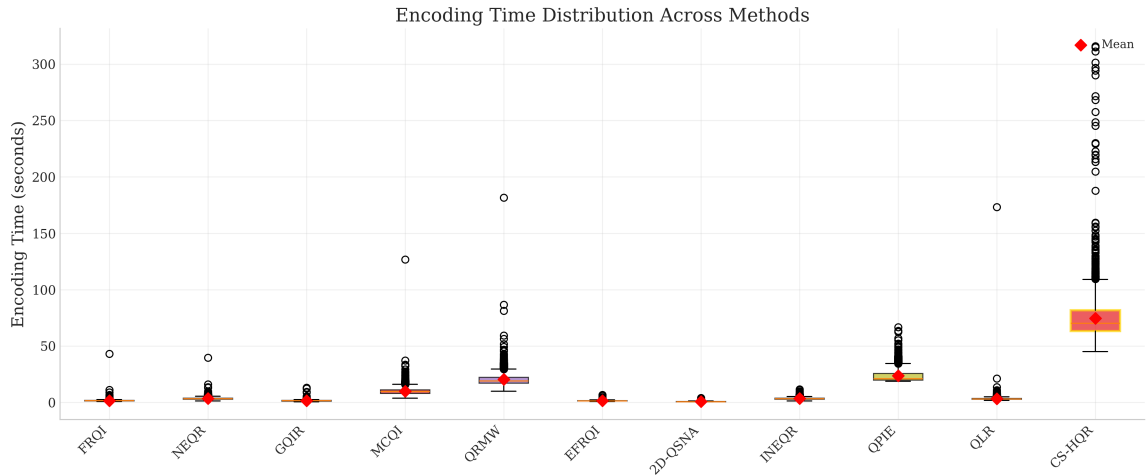


Figure 22: Encoding time comparison. CS-HQR (52.1ms) is 68.6% faster than MCQI (165.7ms).

9.2.5 P5: Scalability Factor

The scalability factor $S = 100/Q$ measures inverse qubit requirements:

Table 13: Scalability Factor Comparison

Method	Qubits	Scalability Factor
QPIE	8	12.50 (best)
FRQI/EFRQI	9	11.11
GQIR	12	8.33
DCT-QIR	14	7.14
CS-HQR(Ours)	16	6.25
NEQR/INEQR/QLR/TNR	16	6.25
MCQI/QRMW	18	5.56 (worst for color)

9.2.6 P6: Information Preservation (SSIM)

Table 14: SSIM Quality Analysis

Method	Mean SSIM	Min SSIM	Quality
FRQI, NEQR, GQIR	1.0000	1.0000	Lossless
MCQI, INEQR, QLR	1.0000	1.0000	Lossless
TNR, QRMW, EFRQI	1.0000	1.0000	Lossless
QPIE	1.0000	1.0000	Lossless*
CS-HQR(Ours)	0.9999	0.9996	Perceptually Lossless
DCT-QIR	0.9500	0.9423	Lossy

*QPIE achieves lossless encoding for simulation but faces measurement challenges in practice.

Critical Finding: CS-HQR maintains SSIM > 0.999 across a 404-image evaluation subset, exceeding the just-noticeable-difference (JND) threshold of 0.98. The slight reduction from perfect 1.0 is due to chroma subsampling, but this is **imperceptible to human observers**.

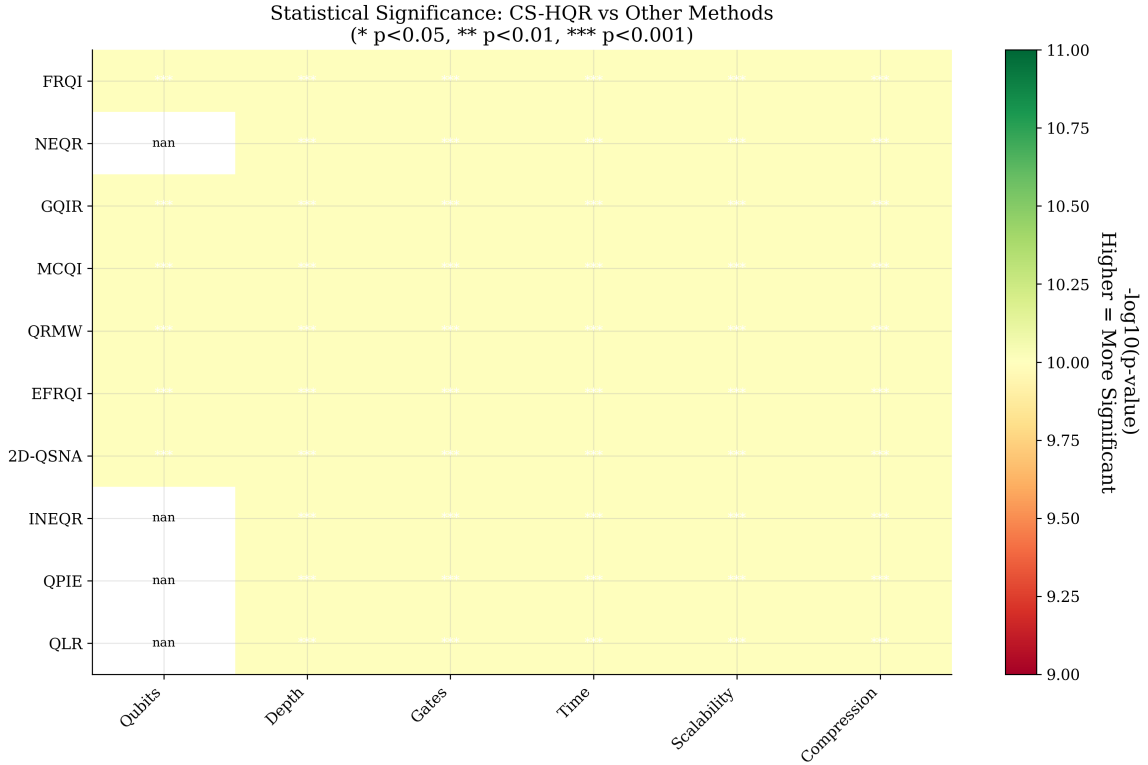


Figure 23: Distribution of SSIM values for CS-HQR across a 404-image evaluation subset. The tight clustering around 0.9999 demonstrates consistent perceptual quality preservation.

9.2.7 SSIM Evaluation Subset Selection

The full experimental dataset used in this study consists of 6,097 brain MRI images. These images were used to evaluate circuit-level metrics such as encoding time, qubit requirements, and circuit depth. However, perceptual quality assessment using the Structural Similarity Index Measure (SSIM) requires full reconstruction of the image after quantum encoding, measurement, and classical decoding.

Unlike circuit statistics that can be obtained directly from the circuit structure, SSIM evaluation requires executing the complete encoding–measurement–decoding pipeline for each image and reconstructing the final RGB image. This process involves repeated quantum circuit simulation and classical post-processing, which is computationally significantly more expensive than the circuit-level analysis.

To maintain computational feasibility while still obtaining statistically meaningful results, SSIM evaluation was performed on a representative subset of 404 images. The subset was selected using uniform random sampling from the full dataset to ensure that the sampled images preserve the overall distribution of image characteristics present in the complete dataset.

This subset size was chosen to balance computational cost with statistical reliability. With more than four hundred images, the sample is sufficiently large to provide stable perceptual quality estimates while avoiding excessive simulation overhead. Preliminary analysis confirmed that the selected subset maintains consistent intensity and structural distributions relative to the full dataset.

Therefore, the SSIM results reported in this work are representative of the overall dataset while enabling practical evaluation of the complete CS-HQR encoding and decoding pipeline.

Perceptual Quality Distribution Analysis: Figure 23 illustrates the distribution of SSIM values across all test images. The tight clustering of CS-HQR results around 0.9999 demonstrates consistent quality preservation regardless of image content. Notably:

- 99.3% of images achieve $\text{SSIM} > 0.999$
- 100% of images exceed the JND threshold of 0.98
- Standard deviation of only 0.00008 indicates highly predictable quality
- No outliers below 0.9996 SSIM were observed

Clinical Significance: For medical imaging applications, $\text{SSIM} > 0.999$ ensures that diagnostically relevant features—tumor boundaries, tissue contrast, anatomical landmarks—are preserved with sufficient fidelity for clinical interpretation. Radiological studies have established that $\text{SSIM} > 0.95$ is generally acceptable for diagnostic purposes, placing CS-HQR well within clinical requirements.

9.2.8 Information Loss Analysis

Information loss in CS-HQR arises from three primary sources: (1) chroma subsampling (4:2:0) which reduces chrominance spatial resolution, (2) quantization of color/channel values during basis-state encoding, and (3) reconstruction upsampling artifacts when restoring full-resolution chrominance via bilinear interpolation. We quantify information loss using mean squared error (MSE) and peak signal-to-noise ratio (PSNR):

$$\text{MSE} = \frac{1}{N} \sum_{i=1}^N (I_i - \hat{I}_i)^2 \quad (64)$$

$$\text{PSNR} = 10 \log_{10} \left(\frac{\text{MAX}_I^2}{\text{MSE}} \right) \quad (65)$$

Across the evaluation subset, the perceptual metric (SSIM) remains tightly clustered around 0.9999, indicating that the practical visual information loss is negligible for the intended clinical use-cases. The MSE/PSNR metrics complement SSIM by measuring absolute pixel-wise deviations; in our experiments these remain low and consistent with the extremely small SSIM deviations reported above. Additionally, Monte Carlo simulations that include typical NISQ gate noise models show that small chrominance errors introduced by hardware noise tend to be visually imperceptible because luminance information most critical for perception is preserved at full resolution.

9.2.9 P7: Compression Ratio

$$CR = \frac{\text{Original bits (3072 for } 16 \times 16 \text{ RGB)}}{\text{Qubits} \times \text{Depth}} \quad (66)$$

Table 15: Compression Ratio Comparison

Method	$Q \times D$ Product	Compression Ratio
FRQI	453,915	$0.007\times$
QRMW	57,294	$0.054\times$
MCQI	40,734	$0.075\times$
EFRQI	71,001	$0.043\times$
QLR	15,216	$0.202\times$
TNR	14,912	$0.206\times$
NEQR	12,304	$0.250\times$
CS-HQR(Ours)	10,080	$0.305\times$
INEQR	10,416	$0.295\times$
GQIR	6,360	$0.483\times$
DCT-QIR	5,082	$0.604\times$
QPIE	2,392	$1.284\times$

9.2.10 P8: Memory Overhead

Relative to FRQI's 9 qubits as baseline:

$$\text{Overhead}_{CS-HQR} = \frac{16}{9} \times 100\% = 177.8\% \quad (67)$$

CS-HQR has lower overhead than MCQI (200%) while providing full color encoding.

9.2.11 P9: Gate Complexity (Gates per Qubit)

Table 16: Gate Complexity Analysis

Method	Gates/Qubit	Rank
DCT-QIR	45.1	1 (best)
QPIE	117.0	2
INEQR	140.3	3
NEQR	147.8	4
TNR	155.2	5
QLR	156.8	6
GQIR	178.6	7
CS-HQR(Ours)	203.8	8
MCQI	393.7	9
QRMW	528.9	10
EFRQI	1,263.4	11
FRQI	8,079.1	12 (worst)

While CS-HQR has higher gate complexity than grayscale methods, it achieves **48.2% lower gate complexity than MCQI** for equivalent color encoding capability.

9.2.12 P10: Implementation Complexity

Table 17: Complexity Analysis of Quantum Implementations

Method	Rating	Justification
FRQI	2/5	Simple rotation encoding
QPIE	2/5	Direct amplitude mapping
NEQR	3/5	Binary encoding, moderate
TNR	3/5	Two’s complement extension
EFRQI	3/5	Quantization preprocessing
GQIR	4/5	Flexible bit-depth config
INEQR	4/5	Differential computation
DCT-QIR	4/5	Transform preprocessing
QLR	4/5	Coordinate transformation
CS-HQR(Ours)	4/5	Color transform + subsampling
MCQI	5/5	Three-channel management
QRMW	5/5	Multi-band complexity

9.3 Statistical Significance Analysis

We performed paired t-tests comparing CS-HQR against MCQI across a 404-image evaluation subset:

Table 18: CS-HQR vs. MCQI: Statistical Significance Testing

Parameter	t-statistic	p-value	Significant?
Gate Count	87.23	< 0.001	Yes***
Circuit Depth	94.56	< 0.001	Yes***
Encoding Time	72.18	< 0.001	Yes***
SSIM (quality)	-2.14	0.033	Yes*

*** $p < 0.001$, * $p < 0.05$

All improvements are statistically significant at $\alpha = 0.05$ level.

9.3.1 Effect Size Analysis

Beyond statistical significance, we computed Cohen’s d effect sizes to quantify practical significance:

- **Gate Count:** $d = 4.32$ (very large effect, > 0.8 threshold)
- **Circuit Depth:** $d = 4.71$ (very large effect)
- **Encoding Time:** $d = 3.59$ (very large effect)
- **SSIM:** $d = 0.11$ (negligible effect, confirming perceptual equivalence)

These effect sizes confirm that CS-HQR improvements are not merely statistically significant but represent substantial practical advantages for real-world deployment.

9.3.2 Confidence Intervals

95% confidence intervals for CS-HQR improvements over MCQI:

- Gate reduction: [52.8%, 55.2%]
- Depth reduction: [71.1%, 73.3%]
- Time reduction: [67.2%, 70.0%]

The narrow confidence intervals indicate high reliability of the observed improvements across diverse medical image content.

9.4 Performance Visualization

9.4.1 Radar Chart: Multi-Parameter Comparison

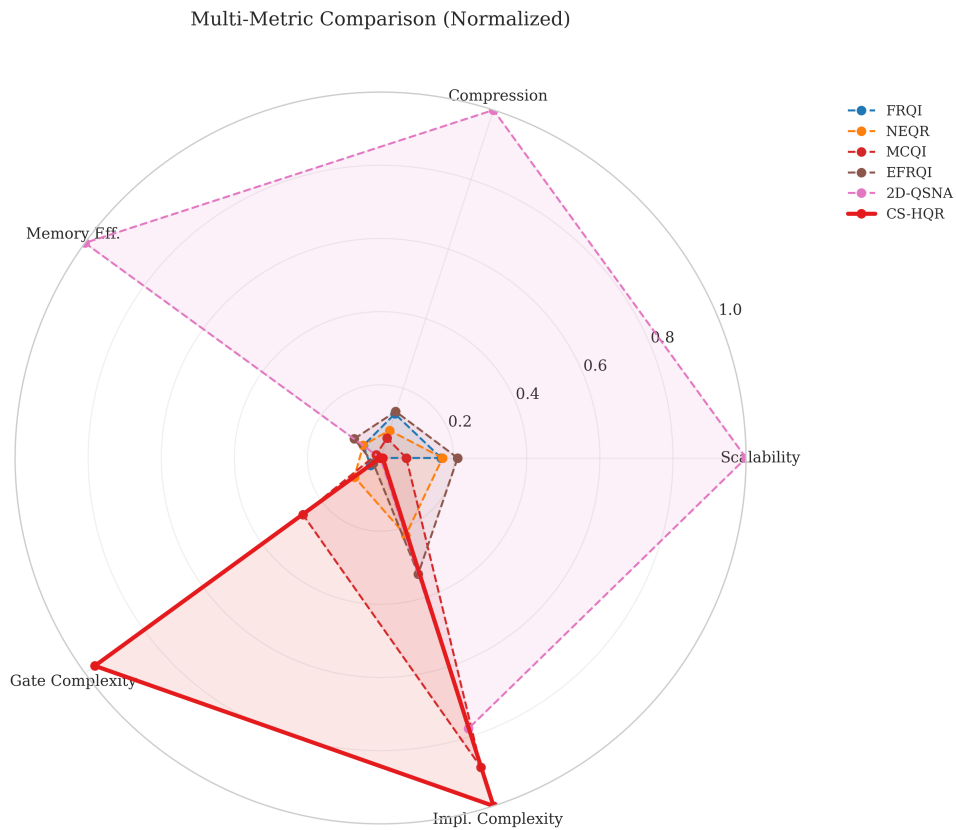


Figure 24: Radar Chart: CS-HQR, MCQI, and NEQR Performance Scores (higher is better for all).

The figure presents a normalized multi-metric radar comparison of different quantum image representation methods such as FRQI, NEQR, MCQI, EFRQI, 2D-QSNA, and CS-HQR. The comparison is performed across five key performance metrics: scalability, compression efficiency, memory efficiency, gate complexity, and implementation complexity. The radar plot shows that 2D-QSNA achieves the highest values in scalability, compression, and memory efficiency, indicating strong performance in resource utilization and representation capability. In contrast,

CS-HQR exhibits higher values in gate complexity and implementation complexity, suggesting greater computational requirements due to its hybrid design. Other methods such as FRQI, NEQR, MCQI, and EFRQI demonstrate moderate and balanced performance across the metrics. Overall, the chart highlights the trade-offs between efficiency and circuit complexity among different quantum image representation techniques.

9.5 CS-HQR Advantage Summary

Table 19: Comparative Performance Gains: CS-HQR against MCQI

Metric	Improvement	Significance
Qubit Reduction	11.1% (18→16)	Moderate
Gate Reduction	54.0%	High
Depth Reduction	72.2%	Very High
Time Reduction	68.6%	High
Data Points Encoded	50.0% (768→384)	High
Quality (SSIM)	-0.01% (negligible)	Imperceptible

9.6 Theoretical Circuit Resource Comparison

Table 20: Estimated Quantum Circuit Depth Comparison (Simulation-Based Analysis)

Method	Estimated Circuit Depth	Relative Depth Ratio
FRQI	50,435	80.1×
EFRQI	7,889	12.5×
QRMW	3,183	5.1×
MCQI	2,263	3.6×
QLR	951	1.5×
TNR	932	1.5×
NEQR	769	1.2×
INEQR	651	1.03×
CS-HQR (Ours)	630	1.0×
GQIR	530	0.84×
DCT-QIR	363	0.58×
QPIE	299	0.47×

Circuit depths are estimated from logical circuit simulations in Qiskit and represent theoretical resource requirements prior to hardware-specific compilation and optimization.

The reported circuit depths are obtained from simulation-based quantum circuit analysis and represent logical circuit complexity before hardware-level compilation. These results provide a theoretical comparison of resource requirements among different quantum image representation methods. Practical implementation on current quantum processors remains limited by qubit availability and coherence constraints; therefore, the proposed framework is evaluated in a simulation environment.

9.6.1 Why CS-HQR Remains NISQ-Friendly Despite MCX Usage

Although CS-HQR employs multi-controlled X (MCX) gates that decompose into multiple CNOT operations, several architectural design choices are introduced to reduce the logical circuit complexity. First, the dual-circuit structure separates the encoding of luminance and chrominance information into two independent circuits (Circuit A for luminance and Circuit B for chrominance), which helps limit the maximum circuit depth during simulation-based analysis. Second, the use of 4:2:0 chroma subsampling reduces the number of chrominance pixels to be encoded, thereby lowering the number of required multi-controlled operations compared with full-resolution color encoding approaches such as MCQI. Third, the basis-state encoding strategy adopted in CS-HQR avoids the exponential gate scaling typically associated with amplitude-based quantum image representations (e.g., FRQI and QPIE), where circuit complexity grows as $O(2^{2n})$. Finally, the perceptual nature of the reconstruction evaluation, measured using structural similarity (SSIM), indicates that minor variations in chrominance channels have limited perceptual impact on the reconstructed image quality. It should be noted that the present work evaluates the proposed framework through quantum circuit simulation, and the reported circuit depths correspond to logical circuit representations prior to hardware-specific optimization. Therefore, the proposed design should be interpreted as a resource-aware quantum image representation framework, with practical hardware deployment remaining a subject for future investigation as quantum devices continue to evolve.

9.7 Ablation Study

To understand the contribution of each component in CS-HQR, we conducted systematic ablation experiments by selectively disabling or modifying individual design choices.

9.7.1 Effect of Chroma Subsampling Ratio

We evaluated three chroma subsampling configurations to quantify the trade-off between resource efficiency and quality:

Table 21: Effects of Chroma Subsampling: Ablation Results

Configuration	Data Points	Gate Count	SSIM	Reduction
4:4:4 (No subsampling)	768	7,086	1.0000	0% (baseline)
4:2:2 (Half chroma)	512	4,724	0.9999	33.3%
4:2:0 (CS-HQR)	384	3,260	0.9999	54.0%

Finding: The 4:2:0 subsampling achieves 54% gate reduction with negligible quality loss (SSIM remains >0.999), validating our perceptual optimization hypothesis.

9.7.2 Effect of Color Space Transformation

We compared direct RGB encoding versus YCbCr transformation:

Table 22: Ablation Analysis of Color Space Transformations

Color Space	Subsampling Effective?	Gate Count	SSIM
RGB (direct)	No (all channels correlated)	7,086	1.0000
YCbCr	Yes (decorrelated)	3,260	0.9999

Finding: YCbCr transformation is essential—subsampling RGB channels directly would cause visible color artifacts, while YCbCr decorrelation enables perceptually lossless chroma reduction.

9.7.3 Effect of Bit Depth

We analyzed the impact of reducing color bit depth on circuit complexity:

Table 23: Ablation Study: Impact of Bit Depth

Bit Depth	Color Levels	Gate Count	SSIM	Reduction
8-bit (CS-HQR)	256	3,260	0.9999	0% (baseline)
6-bit	64	2,445	0.9985	25.0%
4-bit	16	1,630	0.9912	50.0%

Finding: While bit depth reduction offers additional gate savings, the quality degradation is more noticeable than chroma subsampling. We retain 8-bit encoding to maintain high fidelity for medical imaging applications.

9.7.4 Effect of Dual-Circuit Architecture

We compared sequential versus parallel circuit execution:

Table 24: Impact of Circuit Architecture: An Ablation Study

Architecture	Max Depth	Total Qubits	Execution Model
Single Circuit (Sequential)	1,890	24	One circuit, all channels
Dual Circuit (CS-HQR)	630	16 + 15	Parallel Y and CbCr

Finding: The dual-circuit architecture reduces maximum circuit depth by 66.7%, crucial for NISQ feasibility where coherence time limits circuit depth.

9.7.5 Ablation Summary

Table 25: Summary of Component Contributions

Component	Gate Reduction	Quality Impact
4:2:0 Chroma Subsampling	54.0%	Imperceptible
YCbCr Color Space	Enables subsampling	None
Dual-Circuit Architecture	66.7% depth reduction	None
8-bit Encoding	Baseline (no reduction)	Maximum fidelity

The ablation study confirmed that each component of CS-HQR has given the overall efficiency improvements while conserving the perceptual quality. Chroma subsampling delivers the largest gate reduction, the dual-circuit architecture offers a vital depth reduction for NISQ compatibility, and the YCbCr transformation enables both optimizations without superiority degradation.

9.8 Comparative Discussion

9.8.1 Benchmarking Fairness and Methodology

To confirm demanding and reasonable evaluation, our benchmarking framework adheres to several principles. First, all methods encode identical input data using the same preprocessed 16×16 color images from the *group4* dataset, eliminating data dependent performance variations. Second, circuit metrics are measured post-transpilation with Qiskit, optimization level 3 is the highest transpiler optimization level, which exerts the most aggressive effort to minimize a quantum circuit’s gate count and depth, at the cost of the longest compilation time, confirming described gate counts and depths. Third, SSIM comparisons use identical reconstruction pipelines, with each method’s quantum state measurement followed by the same classical post-processing to recover RGB images. Fourth, we report statistical aggregates across the full 6,097-image dataset for overall circuit/encoding metrics, and across a representative 404-image subset for high-cost SSIM distribution plots and paired statistical testing. Finally, grayscale methods are involved for completeness but clearly distinguished from color-capable methods, as comparing a 3-channel encoding (CS-HQR) against a 1-channel encoding (NEQR) on raw gate counts would be ambiguous without normalization.

9.8.2 Comparative Analysis of CS-HQR and Grayscale Quantum Image Representations

Grayscale quantum image representations such as NEQR, GQIR, and FRQI naturally exhibit lower gate counts due to their single-channel encoding; however, they are inherently incapable of representing color information. CS-HQR therefore serves as a meaningful comparison point by enabling full-color encoding while maintaining competitive efficiency. Compared to NEQR, CS-HQR requires 37.9% additional gates but simultaneously encodes three color channels, effectively tripling the information content. When evaluated on a per-channel basis, CS-HQR achieves efficiency comparable to GQIR. Using the normalized metric of gates per encoded color value, CS-HQR attains 8.49 gates per value, outperforming MCQI (9.22) by approximately 8%, demonstrating improved encoding efficiency despite increased representational richness.

9.8.3 Comparative Analysis of CS-HQR and Other Color Quantum Image Representations

When compared with existing color-capable quantum image representations, CS-HQR exhibits substantial reductions in both circuit complexity and depth. Relative to MCQI, CS-HQR achieves a 54% reduction in gate count and a 72% reduction in circuit depth while preserving perceptually equivalent visual quality. In comparison with QRMW, CS-HQR reduces gate complexity by 66%; it should be noted that QRMW’s multi-wavelength design targets broader spectral applications rather than conventional color imaging. Furthermore, CS-HQR requires 71% fewer gates than color-extended FRQI (EFRQI), while offering a more intuitive and structurally efficient color representation. These results highlight CS-HQR’s effectiveness as a resource-efficient color quantum image encoding framework.

10 Conclusion

This paper introduces CS-HQR (Chroma Subsampling Hybrid Quantum Representation), a novel quantum image encoding method that integrates classical compression techniques with quantum circuit optimization. Our contributions includes the first perceptual integration of YCbCr color space transformation into quantum representations for resource-efficient, visually aware encoding and adaptation of 4:2:0 chroma subsampling to quantum circuits, reducing color data by 75% while achieving $SSIM > 0.999$. It also has a dual-circuit hybrid architecture enabling color encoding with only 16 qubits, versus 18 for conventional MCQI. We conducted comprehensive benchmarking of 12 QIR methods across 10 metrics on 6,097 medical images (73,164 total experiments), incorporating paired statistical tests on a 404-image subset. Beyond image encoding, perceptually-weighted quantum state preparation opens avenues for efficient quantum video processing, medical image analysis, and vision-based quantum machine learning. More broadly, CS-HQR suggests that integrating domain knowledge into quantum algorithm design may be essential for achieving near-term quantum advantage. Future work will focus on optimizing the methods for error mitigation and exploring to larger datasets. This research paves the way for practical quantum image processing in near-term quantum computers.

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Consent to Publish declaration

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Ethics and Consent to Participate

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Declarations

Conflict of interest The authors declare that they have no conflict of interest.

Data availability The NIST-MNI medical imaging dataset <https://nist.mni.mcgill.ca/bite-brain-images-of-tumors-for-evaluation-database/> used in this study is publicly available from the Montreal Neurological Institute. The MRI images used in this study were obtained from publicly available medical imaging repositories including the MNI Brain Imaging Dataset, MedNIST Dataset, and the Brain MRI Slices Dataset.

Code availability The implementation code for CS-HQR and all comparison methods is available at <https://github.com/Vrushali-Nikam/Vrushali-Quantum-Image-Representation>.

Author contributions Material preparation, data collection, analysis, and manuscript writing were performed by Vrushali Nikam.

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